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Does corruption have a sanding or greasing impact on innovation? Reconciling the contrasting perspectives through a systematic literature review

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ABSTRACT

There has been a significant increase in the number of academic studies published on the relationship between corruption and innovation in recent years. Various relationships have been conceptualized, including the "sand" and "grease" perspectives which propose that corruption reduces and increases innovation, respectively. In light of interest in the topic showing growing momentum, we review the literature on corruption and innovation for the purpose of reconciling these proposed theoretical perspectives. Following a systematic literature review methodology, we explain the composition of the literature and map the key conceptualizations of the corruption-innovation relationship. We link the variables, measures, and theories employed in the literature to proposed conceptual relationships. Furthermore, we outline the key patterns in which the sanding or greasing impact is more likely to be observed. Based on the systematic review, we propose a research agenda for corruption-innovation research and discuss policy implications.

1. Introduction

Corruption impacts economies worldwide by incentivizing firms to shift their efforts towards unproductive behaviours like bribery and fraud. Its impact on overall productivity is unequivocal, costing the global economy about 5 % of GDP on a yearly basis (United Nations, 2018). However, less clear is the relationship between corruption and innovation. A large stream of literature argues a causal link between corruption and innovation, yet it is debated whether corruption is a deterrent to innovation efforts (Méon and Weill, 2010) or an enabler, a necessary means of facilitating innovation (Cuervo-Cazurra, 2016). Further, some literature depicts corruption and innovation as concomitant phenomena coexisting in certain national contexts (Wegner, 2019), while others regard them as substitutive strategies initiated by decision makers (Burnham et al., 2018).

Researchers have begun to acknowledge the importance of the corruption-innovation relationship, answering calls for further research on the topic (Anokhin and Schulze, 2009). As we go on to show, there has been a 178 % increase in published articles between the 2004–2016 and 2017–2021 time periods. However, the recent, substantial growth in the number of studies has not been met with sufficient focus on synthesizing the literature. A synthesis is especially important because the literature is fragmented across several academic disciplines including management (e.g. Burnham et al., 2018; Xu et al., 2019), economics (e.g. Athanasouli and Goujard, 2015; Nam and Thanh, 2021), and political science (e.g. Belloc, 2014; Wegner, 2019). Furthermore, the contextual focus (e.g. industries or countries) and level of analysis of both corruption and innovation (i.e. firm, region, or country) differs across studies, causing diverging theoretical explanations and inconsistent conclusions to persist.

As a result of this lack of synthesis, additions to the body of research seem to be widening rather than bridging the gap between perspectives. Consequently, relevant theoretical and policymaking questions remain unaddressed. For example, in which circumstances (e.g. institutional environments, firm characteristics, industries) is corruption beneficial or harmful to innovation? What types of innovation are impacted negatively or positively by corruption? Does corruption incline firms towards a short-term innovation strategy as opposed to a long-term innovation strategy? Can we expect anti-corruption policies to have a differential impact across firms and innovation activities?

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Accordingly, we conduct a systematic literature review (SLR) that synthesizes the literature on corruption and innovation, including compatible, contradictory, and counter-intuitive findings. Our primary goal is to gain a comprehensive understanding of the conceptual frameworks linking corruption to innovation, the strength and directionality of the relationships, and the variables and theoretical mechanisms channeling those relationships. A more accurate understanding of the theoretical underpinnings, boundary conditions, and contextual specificities of the corruption-innovation relationship can generate insights that improve policymaking. This is imperative since corruption's cost to the global economy means that it cannot exist unabated even if institutional conditions render it effective for promoting innovation.

Our SLR consists of two interrelated analyses. The first is a conceptual mapping of the various ways that the relationship between corruption and innovation has been modelled. This analysis helps us identify the dominant conceptualization in the literature as well as alternative perspectives put forth by scholars. The second is an antecedent-moderator-consequence analysis that delves deeper into the causal link between corruption and innovation—the primary relationship identified in the conceptual mapping. In this analysis, we further decompose the antecedents, consequences, contextual factors, and theoretical perspectives used in the literature and explain their linkage to causal conceptualizations of the corruption-innovation relationship. Along with our dual analyses, we synthesize the findings of our review into patterns of factors that underlie a negative (sanding) and positive (greasing) causal relationship between corruption and innovation. Based on the results of our SLR, we discuss implications for theory and policy, as well as research directions for a more complete understanding of the corruption-innovation relationship.

2. Corruption and innovation: theoretical perspectives and the need for synthesis

The decision to innovate is contingent on the degree to which firms can appropriate returns from their innovation (Harabi, 1995). One of the determining factors of innovation appropriability is the institutional environment, which encompasses the rule of law, judiciary, and regulatory system. By supporting contract enforcement, assuring the receipt of government resources such as intellectual property rights (IPR), and providing transparency regarding regulatory procedures (Alam et al., 2019), strong institutional environments increase the returns from the development and commercialization of innovation.

Corruption affects the decision to innovate. Defined as the misuse of public power for personal benefit (Rodriguez et al., 2005), corruption occurs between firms and government officials and typically entails practices such as bribery, kickbacks in public procurement, and the embezzlement of property (Jensen et al., 2010). The literature has emphasized two, seemingly contradictory perspectives regarding corruption's impact on firm innovation, spurring a debate that spans across academic disciplines on whether corruption acts as sand or grease (e.g. Krammer, 2019; Heo et al., 2021).

The sanding perspective argues that corruption negatively impacts innovation by imposing additional costs and raising uncertainty. Firms are faced with a lower level of appropriability since government resources required for innovating, such as licenses, permits and IPR (and their enforcement) may require additional and often unforeseen payments. Thus, corruption acts as an additional tax (Bardhan, 2017; Rose-Ackerman, 1999), deterring firms from pursuing innovation that would otherwise be profitable.

The greasing perspective suggests that corruption can have a positive impact on innovation by reducing transactional uncertainty (Anokhin and Schulze, 2009). Within a weak institutional context marked by red tape and an ineffective legal system, corruption either accelerates or enables exchanges that would have stalled or not have taken place (Cuervo-Cazurra, 2016). Corruption thus fosters a sense of order in the business environment, such that the added costs of corruption are offset by reduced uncertainty. For the innovating firm, the knowledge of how much to bribe and to whom for obtaining a license may be superior to the uncertainty of not knowing if and when a license will be granted. A more advantageous form of corruption occurs in captured institutions, a situation when the state can exercise its power to provide preferential treatment (Dorobantu et al., 2017). Firms in a position to pay bribes obtain government resources to the exclusion of their competitors, creating monopolistic advantages for firms (Kurer, 1993).

The growth in the literature on the corruption-innovation relationship has not been met with sufficient conceptual development focused on coherently synthesizing these perspectives. Baseline findings that purport either a sanding or greasing relationship have accumulated with little acknowledgement of the interdependency between the two perspectives. Inconsistent findings may be driven by a multiplicity of variable measurements, contextual and firm-level differences, or various disciplinary approaches across studies. A synthesis of the literature can foster a better understanding of the measures, mechanisms, and boundary conditions that determine the strength and directionality of the relationship. Moreover, future researchers can use this foundation of knowledge to move beyond conceptualizing the relationship as a dichotomy to more nuanced questions regarding either corruption's sanding or greasing impact, their possible co-existence, and other possible non-causal conceptualizations that we go on to uncover through our SLR. Political scientists and public administration scholars, for example, conceptualize weak national innovation capacity and high levels of corruption as loosely-coupled social-cultural artifacts of post-socialist countries (Švarc, 2017). Strategy researchers, on the other hand, view bribery and innovation activities as strategic options contingent on a firm's external and internal environment (Burnham et al., 2018; Xu and Wang, 2019).

Since the fundamental issues regarding the causality, directionality, and theoretical mechanisms of corruption-innovation relationship remain unaddressed, broader insights into relevant issues, such as normative and policymaking implications of corruption's greasing impact, have not been brought to light. The negative consequences of corruption to society are well established, contributing to societal problems such as deficient infrastructure, poverty, and inequality (Kaufmann, 1997; Moran, 2001). A critical and in-depth examination of corruption's potential greasing impact can move us past the simplistic, consequentialist conclusion that corruption is acceptable and help build a normative case against corruption as a means of facilitating innovation.

A nuanced understanding of the corruption-innovation relationship is also imperative given the importance of innovation as a policymaking concern. For instance, current and emerging information and communication technologies (ICT) offer significant promise of transformation and development of low- and middle-income countries, where corruption tends to be rife (Kaplinsky and Kraemer-Mbula, 2022). In recent years, there has been considerable emphasis on supporting innovative start-ups through a broader set of policies that encompass both infrastructure and institutions (Audretsch et al., 2020). A comprehensive understanding of how the corruption-innovation relationship differs across types of firms, industries, and innovation efforts can inform anti-corruption policy that is attuned with these innovation objectives.

3. Methodology

Following the research of Hurmelinna-Laukkanen and Yang (2022), Rotolo et al. (2022), Stornelli et al. (2021), and the outline proposed by Tranfield et al. (2003), we carried out a systematic literature review to examine prior research and synthesize results. A SLR employs an article search strategy with well-chosen keywords to ensure a comprehensive sample and employs strict inclusion and exclusion criteria to ensure the relevance of studies. SLR also uses a pre-developed template for data entry with well-defined categories and coding schemes drawn from the prior literature. Compared with a traditional literature review, which

uses a subjective approach to summarize literature and distill ideas from published research (Wakefield, 2014), a SLR ensures less biased results (Kitchenham et al., 2009). We also conducted meta-analyses on subsets of the data to provide further justification for some of our arguments. We now explain the process used to identify, code, and analyze the relevant articles.

3.1. Article identification and selection

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach was used to address the challenge of identifying relevant studies and summarizing findings across a wide-ranging literature (Moher et al., 2009). PRISMA is a set of elements for reporting systematic reviews and meta-analyses that are supported by evidence. It was created to set better standards for openness and clarity in methodology. PRISMA offers a graphic representation of the identification, screening, eligibility, and inclusion of studies together with the justifications for their exclusion at every level of the review process.

The article identification stage consisted of a thorough keyword search through the Web of Science database. After an extensive review of the literature, two sets of keywords were identified for each of corruption and innovation. The keywords used for corruption included "Corruption", "Rent-seeking", "Institutional capture", "Kick back", "Bribery", "Side payments", "Greasing the wheels", "Elite capture", and "gift giving." The keywords used for innovation included "Innovation", "R&D", "Research and Development", "Research & Development", "Patent", and "Product development". Fifty-four different permutations of corruption and innovation keywords were applied to titles, abstracts, or author-supplied keywords to identify relevant English language, peer-reviewed scholarly articles. A total of 1067 articles spanning the 17-year period between 2004 and 2021 were found. We also used a number of inclusion and exclusion criteria in order to produce a sample with acceptable proportions. After excluding foreign language articles, conference proceedings and book chapters, 772 articles were retained. The bibliographic information of all 772 articles (title, journal name, publication year, abstract, authors, keywords) was exported to Excel. Each article was assigned a unique ID which did not change in all subsequent analyses. Following the PRISMA framework, the choice of which articles to include was made after reading the eligible papers' titles, abstracts, and complete text, if needed. Only articles with a key focus on corruption and innovation rather than a casual mention of either concept were included. 669 articles were removed at this stage, leaving a final sample of 103 articles. Fig. 1 provides an overview of the article selection process.

3.2. Coding

A systematic coding process was completed as part of a content analysis to extract important information from each article (see Grodal et al., 2021). The information in a content analysis is both manifest and latent. Manifest content, such as sample size, is easily observable and can be documented with a high degree of reliability. Contrarily, latent content such as the conceptual relationship between corruption and innovation assumed in a study must be understood using the researcher's judgement because it is implicit and integrated into the message (Holsti, 1969).

3.2.1. Descriptive characteristics of studies

To analyze the methodological approach of the studies, a type of manifest content, the working definitions for all categories of data—for example, empirical studies, conceptual studies, primary data, and

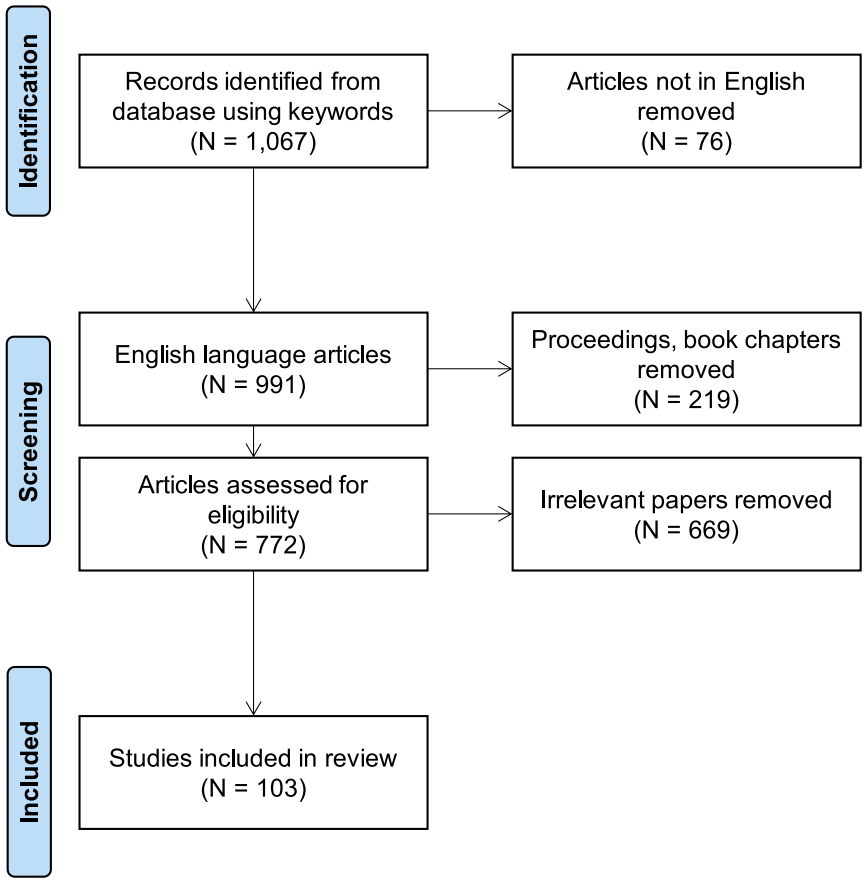

Fig. 1. Article selection process PRISMA flow diagram.

secondary data—were first established. A study was classified as empirical if it collected numerical data and analyzed it using a statistical method. Conceptual studies were classified if they did not contain any data. Primary data refers to data collected from respondents by researchers of a study, whereas secondary data is borrowed from already published sources. Data collection countries, unit of analysis, and empirical approach were all coded. A coding template was created by the research team. A research assistant with prior coding experience was trained to extract methodological data for this project. The coding scheme, a list of data collection categories, and relevant definitions was shared with the research assistant to ensure consistency. Ten articles at a time were provided to the research assistant for data input. The coder read the methodology sections of each study and copied and pasted the exact methodological information without changing any words. As a reliability check, a co-author with specialization in research methods checked 100 % of the data once it had been inputted. See Appendix A for the list of codes.

3.2.2. Variable analysis (antecedent, consequence, moderator, operationalization)

Leveraging pre-existing frameworks helps organize a SLR and increase its reliability (Paul and Barari, 2022). An antecedent-consequence framework has been widely used in SLRs across various disciplines such as information systems (Bhatti et al., 2021; Schlackl et al., 2022), marketing (Mostafa and Kasamani, 2022; Prentice et al., 2022; Sorescu and Schreier, 2021), and operations management (Yang et al., 2021). An antecedent is a variable that occurs before a dependent variable, whereas a consequence could be a final outcome or dependent variable. Moderating variables can detect common trends and inconsistencies in the literature, and are also included in the analysis.

The coding process for this analysis focused on interpreting latent content that required the judgement and prior experience of the researchers. Two of the authors in the paper were responsible for this coding process, which was undergone in three steps. Step one consisted of individually transferring the pertinent data from each article onto the coding procedure form in Excel. The researchers carefully read each study before copying and pasting the most appropriate words from the study that represented the variables of interest. In the second step, the researchers reviewed one another's work to make sure all coding processes were correct and accurate. Step three consisted of inter-coder reliability of the coding protocol. Intercoder agreement was between 89 % and 100 %. Any discrepancies in the coded data were resolved through further discussion.

We used a four-step process to inductively progress from low-level codes towards high level categories. First-order codes are observable objects or concepts found in the articles. Higher level categories are the more abstract constructs that encompass observable concepts. First, each study was thoroughly read and variables were extracted from the hypotheses, conceptual frameworks, and analysis tables. Corresponding to each study, the variables were entered in separate Excel columns to distinguish antecedents, consequences, and moderators. The exact words used by the authors of a study were entered to minimize bias. Second, similar words used to describe the same variable or concept were reconciled to consolidate the set of variables. For example, corruption was represented by several keywords such as bribery, irregular payments, control of corruption, and rent-seeking. The variables were categorized as being associated with corruption, innovation, or other relevant categories. Third, the operationalization of corruption and innovation measures was extracted from each study. The measurements of corruption and innovation were categorized into a higher level of abstraction such as corruption or innovation at the firm, industry, or geographic level (e.g. country, state, or province). In addition, the corruption measures were coded as objective or subjective and innovation measures were coded as input or output to facilitate the planned analysis. The description and examples of measures were entered in a separate column. Fourth, for each moderator, the arguments provided by the study were entered along with a description or example. All measurements and operationalizations were used later to map the conceptual relationships between corruption and innovation and address the inconsistent findings in extant literature. See Appendix B for an illustration of the process.

3.2.3. Conceptual mapping and hypotheses analysis

For the coding underlying the conceptualization of the corruption-innovation relationship, we did not prescribe the type, name, or directionality of relationships a priori. Our objective was to allow the relationships to emerge through our analysis. However, some types of relationship classifications were anticipated based on the knowledge of the researchers. These include the following classifications: unrelated, conceptually related, empirically correlated, substitutive, complementary, and causal. We define the relationship between corruption and innovation as unrelated if no conceptual relationship is discussed. Two variables are conceptually related if the connection between corruption and innovation is discussed. Further, if the researchers also employ data to examine the statistical association between conceptually related variables and find empirical evidence, we defined variables as empirically correlated. Among all empirically correlated relationships, substitutive and complementary are two common types of statistical correlation. A substitutive relationship is one in which the increase in one variable results in a reduction of the other. A complementary relationship is one in which an increase in one variable leads to the increase of the other. Lastly, a causal relationship is the strongest form of statistical correlation between variables—when the impact on a variable is the direct cause of the other, and the cause occurs before the effect.

After the coders reached a consensus regarding the definitions of relationships, they classified the relationships in all 103 articles, discussed the coding process, and resolved any inconsistencies. Not all codes were used in the final analysis. Rather, four major types of conceptual relationships emerged in extant literature that became the focus of this paper: causal, reversed-causal, substitutive, and symbiotic. While similar to the "conceptually related" classification defined earlier, we used the term symbiotic to better reflect the relevance of context in most studies. In ecological research, a symbiotic relationship is a close, prolonged association between species in an ecosystem. We use this analogy to refer to a conceptual relationship between variables deeply embedded in a broader social-economic environment. We further explain these conceptual relationships in the analyses section.

For studies that advocate a causal relationship between corruption and innovation, we checked the directionality of relationships predicted. We anticipated that there would be no more than six types of results: not correlated (i.e. a null hypothesis is supported), correlated (i.e. a null hypothesis is rejected), negative, positive, quadratic (U-shaped or inverted-U shaped), and cubic (S-curved). Our plan was to identify the dominant perspective in the literature and identify the studies that propose different perspectives. After the coding process, we found only three types of relationships between corruption and innovation: correlated, positive, and negative. Our classification of directionality was modified accordingly. Further details are provided in the analyses section.

4. Analyses

Section 4 presents analyses produced from the SLR. It begins with an overview of the literature (Section 4.1) and measurements of variables (Section 4.2), continues with the conceptual mapping of the corruption-innovation relationship that emerged from the SLR (Section 4.3), and finishes with an in-depth analysis of the causality claim between corruption and innovation (Section 4.4).

4.1. Overview of the literature

Several findings explain the evolution of the literature and its current

composition. We observe an exponential rise in the number of published articles on the topic in recent years, such that we divide the literature into two periods (Fig. 2). Of the 103 articles in this review, the vast majority (74 %; 76 articles) were published in the five-year period between 2017 and 2021. From 2004 to 2016, a thirteen-year span, only 27 articles were published (26 %). The growing interest not only necessitates a SLR but indicates that the corruption-innovation relationship will be a relevant topic in the future.

Appendix D details the methodological approach of the literature. The vast majority of articles published are empirical (89 %) rather than conceptual (11 % use no data). This empirical focus has been consistent across both time periods (2004–2016 and 2017–2021). Most studies use quantitative data (88 %) and secondary data sources (97 %). Only a single study uses qualitative data (1 %) and only 3 articles (3 %) relied on primary data. Regarding country-context, there is an even-split between single (50 %) and multi-country (50 %) studies. The rise in the proportion of single country studies during 2017–2021 can be largely attributed to the growing interest in China's anti-corruption enforcement efforts that started in 2012. A consequence of China's focus on anti-corruption has been the availability of corruption related data (e.g. Chen and Yoon, 2019; Yu et al., 2019). Likely for the same reason, studies on China are the most represented among single-country studies (29 %). For multi-country studies, researchers tend to use data from the World Bank Enterprise Surveys (WBES), a global firm-level survey conducted by the World Bank since the 1990s.

4.2. Variable measurements

4.2.1. Corruption measures

Corruption is measured at both the *firm* and *aggregate* levels of analysis (i.e. country or state/region). Measures of corruption can be *subjective*, either of individual or aggregate opinions about the prevalence and impact of corruption, or *objective*, such as participation in bribery or informal payments made to government officials. Tabulating across these two dimensions, we can classify corruption measurements into four types: subjective firm-level, objective firm-level, subjective aggregate-level, and objective aggregate level measures (Appendix E).

Subjective firm-level measures are based on managerial perceptions of corruption in the environment (e.g. Barasa, 2018; Goedhuys et al., 2016). Due to the sensitivity regarding the topic of bribery, projective questions inquiring about informal payments of comparable firms are common. Projective measures are mostly drawn from the WBES.

Objective firm-level measures are typically based on a firm's involvement in bribery or informal payments to government officials. Objective measures include executive's self-responses to questions regarding bribe payments (Ha et al., 2022; Wellalage and Thrikawala, 2021). Fewer studies employ indirect measurements that are not self-responses. For example, some researchers record disclosed administrative and entertainment expenses (Chen and Yoon, 2019; Fu and Jian, 2021), thought to be a means of disguising corruption.

Subjective aggregate-level measures tend to be corruption indices formulated from expert opinions regarding the extent to which corruption is an obstacle to conducting business in a country (e.g. Rocha et al., 2019; Ervits and Zmuda, 2018). Commonly used indices are the Corruption Perceptions Index (CPI) (Transparency International) and the Control of Corruption index (World Bank). In addition, some researchers aggregate data from large-scale, firm-level surveys to measure the overall level of corruption in a region or an industry (e.g. T.V. Nguyen et al., 2016; Rodríguez-Pose and Zhang, 2020). There are few instances of scholars creating their own corruption index (Liu et al., 2021; González et al., 2020).

Objective aggregate measures include self-responses of managers regarding bribery (e.g. De Waldemar, 2012; Paunov, 2016) aggregated to the country or state-level. Several studies employ state or regional-level corruption convictions data from the US (Dincer, 2019), China (e.g. Ren et al., 2021; Xu and Yano, 2017), Spain, and India (Jain, 2021). Additional examples of the four types of measurements can be found in Appendix E.

4.2.2. Innovation measures

Our SLR suggests that innovation is measured by either inputs or outputs at either the firm or aggregate (i.e. country or state/region) levels. Based on these two dimensions, four types of innovation measures emerge: firm-level inputs, firm-level outputs, aggregate-level inputs, aggregate-level outputs (Appendix F).

Firm-level innovation inputs are usually self-reported R&D investment, measured as R&D expenditure as a percentage of sales (e.g. Alam et al., 2019; Sena et al., 2018), or an indicator of whether a firm engaged in R&D over a period of time (e.g. Sivak et al., 2011; Jain, 2021).

Firm-level innovation outputs are more commonly used measures than firm-level innovation inputs. Outputs include patent applications (e.g. Kong et al., 2020; Guo et al., 2021), and patents granted (e.g. Lin et al., 2021; Wu and Dong, 2021) as a measure of volume, and patent citations as a measure of innovation quality (e.g. Lin et al., 2021; Liu et al., 2021). Some researchers use both firm-level input and output together as a holistic measure of innovation activity (e.g. Athanasouli and Goujard, 2015; Xu and Yano, 2017).

Aggregate-innovation outputs at the country or region level tend to be based on patent application data (e.g. number of patents per capita, total number of patents) (e.g. Rosenberg and Tarasenko, 2020; Wen et al., 2020), usually drawn from the World Intellectual Property Organization (WIPO). Less common measures at the country-level are the number of scientific journal articles (e.g. Gimenez and Sanau, 2007) and the proportion of high-tech exports (Allard and Williams, 2020).

Aggregate-level innovation inputs are rarely used. One study employs a measure based on an executive opinion survey regarding the abundance of innovation inputs in a country (Allard and Williams, 2020). Aggregate measures of innovation input tend to be combined with aggregate measures of innovation output to capture the overall innovativeness of a country or region (e.g. Woodside et al., 2016; Mthanti and Ojah, 2018). Additional innovation measurements can be found in Appendix F.

4.3. Conceptual mapping of the corruption-innovation relationship

Although we anticipated a wider range of relationships before conducting the SLR, four primary conceptualizations of the corruption-innovation relationship were found (see Fig. 3). The most prevalent framework, which we label as *causal* (Framework 1), conceptualizes corruption as the antecedent and innovation as the consequence. Of the 103 articles examined, 70 articles theorize or empirically test the effect of corruption on innovation (pathway *a* in Fig. 3). In accordance with the logic in the causal framework, some researchers also investigate the broader economic outcomes to a country or firm as the ultimate consequence of the corruption-innovation relationship. Acknowledging the direct influence of corruption on innovation, fourteen articles investigate the effect of corruption on country or firm growth directly (pathway *c*) or indirectly via innovation (pathways *a* and *b*).

Researchers have conceptualized corruption and innovation as substitutes for achieving firm growth. 12 (11.8 %) articles fall within the *substitutive* framework (Framework 2). For example, R&D and bribery are considered opposing strategies contingent on performance. When performance rises above aspirations, firms engage in long-term activities such as R&D to sustain their performance. When performance falls below aspirations, firms resort to activities considered short-term such as bribery (Burnham et al., 2018; Xu et al., 2019). In this stream of literature, there is no causality between corruption and innovation. Rather, firms choose either corrupt or innovative activities given their assessment of external environmental conditions and firm-level resources.

The next conceptualization characterizes a "*symbiotic*" relationship between corruption and innovation (Framework 3). Similar to the substitutive relationship, a symbiotic relationship does not focus on the

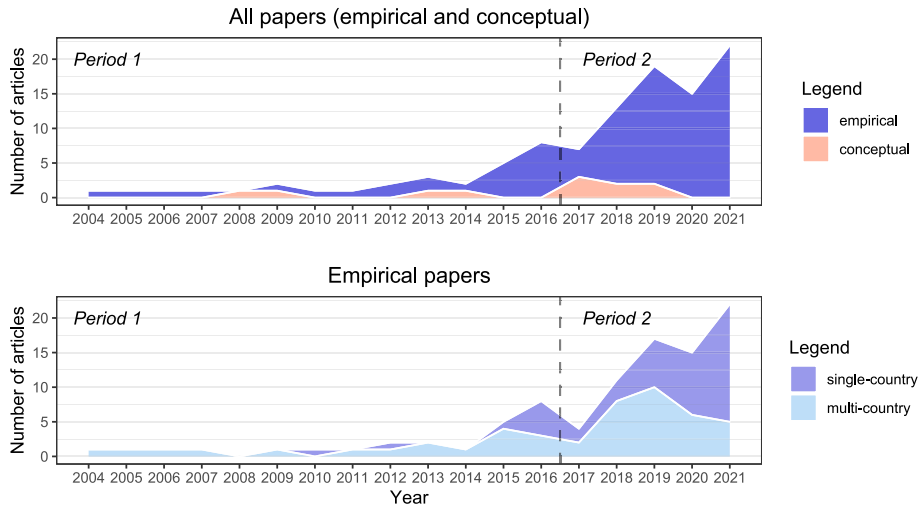

Fig. 2. Number of studies by year.

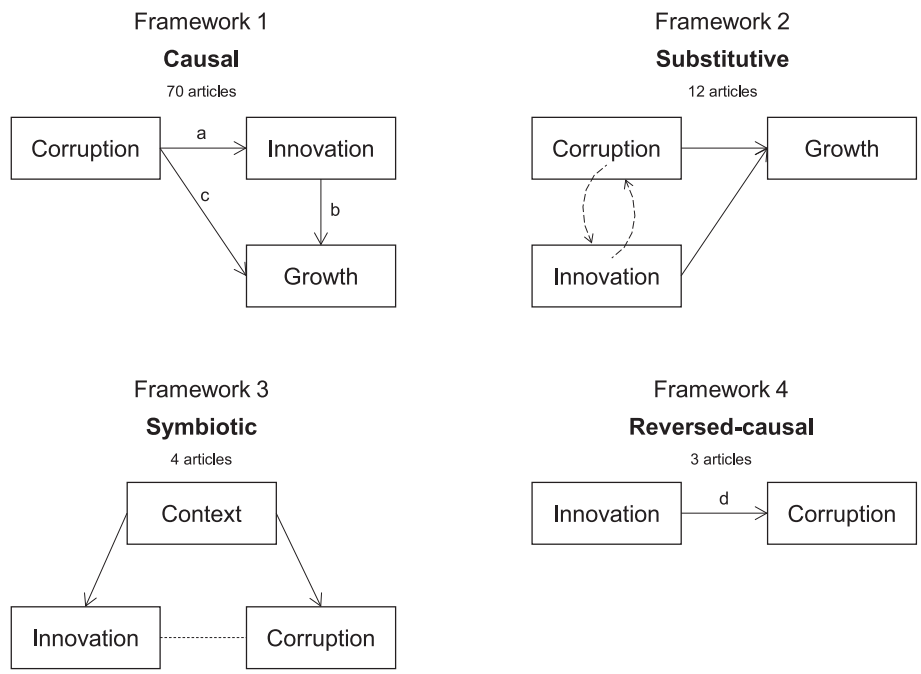

Fig. 3. Conceptual relationships in the corruption-innovation literature.
Note: The labels ("corruption", "innovation", and "growth") used in the figure can denote both country- and firm-level variables. In addition to the 89 articles above, there are also 14 articles that cannot be categorized into any framework. They typically include corruption and innovation variables in their model but do not focus on the relationship between corruption and innovation (e.g. some use corruption as an instrumental variable to investigate other relationships).

sequential order between corruption and innovation. Instead, corruption and innovation are seen as concomitant behaviours driven by the institutional infrastructure of a country. It is a much smaller stream of literature with only four articles. For example, Wegner (2019) argues that in autocratic regimes (e.g. transition economies, post-communist countries), entrepreneurship is channeled by autocratic rulers for the purpose of upholding the regime. The symbiotic framework is typically employed in qualitative research to grasp the interconnected political and economic dynamics in transition and emerging economies.

Lastly, the three articles that examine the relationship in the opposite direction—the impact of innovation on corruption—fall under the *reversed-causal* framework (Framework 4). For example, Ayyagari et al. (2011) find that innovative firms pay more bribes than non-innovators on account of greater interaction with government officials (e.g. applying for a permit and registering a new trademark). Burnham et al. (2018) puts forth the opposing perspective that less innovative firms are more likely to participate in corruption.

In the next section, we delve into the most prevalent research stream, the *causal* framework (Framework 1) to investigate directionality and strength of causal claim in the literature.

4.4. *On the causality between corruption and innovation*

4.4.1. *The dominant argument*

The majority of papers predict that corruption *negatively* affects innovation (see Table 1). The relationship is examined at either the country or firm-level, with 36 out of the 70 articles making this proposition. Among the 36 articles, 16 studies explicitly state a negative hypothesis. The other 20 studies predict a negative relationship but do not include hypotheses. 24 out of 36 studies confirm a negative

Table 1
Summarizing the causal impact of corruption on innovation across studies.

Predicted relationship	Frequency	Literature ID
The dominant perspective		
• Negative hypothesis is proposed	16	91, 286, 330, 386, 1005, 1012, 1058, 1079, 1083, 1092, 1094, 1154, 1159, 1811, 1208, and 1308
• No hypothesis but a negative relationship is explained	20	67, 83, 98, 110, 112, 163, 197, 278, 288, 342, 361, 368, 410, 550, 552, 1051, 1077, 1182, 1197, and 1401
Alternate argument 1		
• Negative and positive hypotheses are proposed	7	12, 135, 211, 227, 1001, 1007, and 1113
• No hypothesis but both directions are explained	4	41, 142, 226, and 611
Alternate argument 2		
• A positive hypothesis is proposed	7	20, 59, 103, 109, 113, 1063, and 1231
• No hypothesis but a positive hypothesis is explained	1	1038
Alternate argument 3		
• No hypothesis proposed and no clear argumentation	14	78, 81, 84, 148, 181, 238, 282, 1002, 1019, 1028, 1032, 1067, 1068, and 1112
Other		
• A null hypothesis is proposed	1	340

relationship, and this number increases to 32 when including studies without hypotheses. A meta-analysis of a subset country-level studies provides further evidence for dominant perspective that corruption negatively affects innovation.[1]

4.4.2. *The alternate arguments*

4.4.2.1. Alternate argument 1: "corruption is both sand and grease". 11 articles in our database make a seemingly incompatible argument that corruption can positively and negatively affect innovation. Researchers embracing *alternate argument 1* take three approaches to straddle the contradictory perspectives. First, their data contains significant within-sample variation such that different results can be observed when researchers zoom in to a particular subset of the data. The within-sample variation includes the characteristics of the firms (e.g. large or small, SOE or non-SOE, domestic firms or MNEs), the institutional context (e.g. developing or developed countries), the nature of corruption (petty or grand), or the types of innovation concerned. Second, researchers incorporate a moderator into the model such that the relationship between corruption and innovation changes in magnitude and sometimes reverses in direction. For example, Goedhuys et al. (2016) suggest that the benefits of bribery only exist in weak institutional environments and that its effects are otherwise detrimental. Last, they propose competing hypotheses based on different theoretical perspectives. For example, T. V. Nguyen et al. (2016) employs transaction cost theory to argue that petty corruption is positively associated with innovation (measured by firm growth variables). On the other hand, they use the resource-based view to argue that corruption negatively affects a firm's innovation (measured by efficiency variables).

[1] In the meta-analysis, country-level corruption was the independent variable, measured using World Bank's Control of Corruption measure. The dependent variable was the number of patents applied for or granted at the country-level. Across a total of 4 studies and 2424 observations, we find that corruption at the country-level has an adverse effect on country-level innovation (pooled effect: 0.737, $p < 0.05$), but heterogeneity exists ($Q = 267.948$, df $= 3; 9$, $p < 0.000$). Note that the Control of Corruption index measure is scaled such that higher values signify lower levels of corruption. As such, the positive effect found indicates a negative relationship between corruption and innovation.

4.4.2.2. Alternate argument 2: "corruption greases the wheels". 8 articles suggest that corruption positively affects innovation. However, the positive effect of corruption seems conditional. To reap the benefits of corruption, firms either operate in a particular institutional context (e.g. Krammer, 2019; Xie et al., 2019) or possess certain firm characteristics (e.g. Tomaszewski, 2018; Xia et al., 2018). The key difference between *alternate arguments 1* and *2* is that articles under the former explicitly model the macro- or firm-level factors that can reverse the directionality of the corruption-innovation relationship. In this sense, a nonmonotonic relationship between corruption and innovation is observed. On the other hand, articles in *alternate argument 2* are situated specifically in contexts in which corruption positively affects innovation. Thus, the relationship is modelled as monotonic. We will discuss country (Section 5.3.1) and firm (Section 5.3.3) characteristics that underlie positive results later in the paper.

4.4.2.3. Alternate argument 3: "no prediction". 15 studies make no empirical prediction regarding the corruption-innovation relationship. Typically, both the sand and grease arguments are presented followed by empirical tests. Dincer (2019), for example, examines the corruption-innovation link using U.S. state-level panel data and confirms that corruption reduces innovation in the long-run. Lacking in theoretical deduction, the contribution of these papers is limited to novel datasets or replication of previous findings in a new context.

5. Making sense of the debate

We now attempt to reconcile the debate regarding corruption's causal impact on innovation by developing an antecedent-moderator-consequence analysis. Our focus specifically is on the corruption-related antecedents, innovation-related consequences, and the moderators (contextual and firm-level) of the corruption-innovation relationship. Furthermore, we overview the theoretical perspective used in the literature. The objective of this section is to explain the contingencies that determine the causality of the corruption-innovation relationship. Accordingly, we highlight linkages between the various causal arguments identified in section 4.4 and antecedents, consequences, moderators, and theoretical perspectives.

5.1. *Corruption and innovation: measurement issues*

5.1.1. *Aggregate versus firm-level corruption measures*

Whether corruption is found to increase or decrease innovation is contingent on the level of the corruption measure, our key antecedent variable. We find that in all 36 articles that formulate a negative relationship between corruption and innovation (see Table 1), most articles (26) test the hypothesis using aggregate corruption measurements. Fewer articles (9) employ firm-level corruption measures while only a single article includes no empirical analysis. The dominant use of aggregate measures seems to reflect the consensus that a country's level of corruption leads to net welfare loss of the entire economy. While bribery allows some firms to benefit by overcoming bureaucratic obstacles and hedging against political risks (Krammer, 2019), the majority of firms are left worse off (Wei, 2000). On the other hand, the greasing hypothesis is more strongly associated with firm-level corruption measures. In all 8 articles that argue a positive relationship between corruption and innovation (*alternate argument 2*), all but one measures corruption at the firm-level. The sole paper that uses an aggregate measure is a cross-country study that includes advanced economies such as Denmark and Canada. A meta-analysis of a subset of firm-level studies provides further evidence of the association between the greasing

hypotheses and firm-level analysis.[2]

5.1.2. Objectivity of corruption measures

In addition, there is a connection between the level of objectivity of the corruption measure and whether corruption acts as a sand or grease in its impact on innovation. This connection is apparent in a cross-tabulation of the corruption measures used and their association with different perspectives of the corruption-innovation relationship (Table 2). There is a high level of association between subjective corruption measures and the claim of no clear relationship between corruption and innovation (i.e. *alternate argument 3*). It may be that the speculative nature of these questions do not accurately reflect the level of corruption in the environment. In addition, objective corruption measures are associated with a potential (i.e. *alternate argument 1*) and definitive (i.e. *alternate argument 2*) greasing impact of corruption. This might be because objective measures of corruption tend to be measured at the firm-level (this is the case in 9 of 11 studies that use objective measures of corruption). Although firm-level measures may capture the benefits a firm obtains by engaging in corruption, they do not reflect the negative impact that corruption in the environment has on firms more broadly. On the other hand, it appears that the sanding perspective is not sensitive to the level of objectivity of corruption measures used. For the majority of articles that propose the sanding perspective, the percentage of papers that use objective and subjective measures are similar (44.4 % versus 58.3 %). The specificity of objective corruption measures may mean they are capturing bribery activity that benefit the engaging firm as compared to more subjective measures that tend to assess corrupt environments more generally.

5.1.3. Level and granularity of innovation activities

Our analysis suggests that the level of the innovation variable impacts whether the study supports the sand or grease argument. Of the studies that propose a positive relationship between corruption and innovation (*alternate argument 2*), *all* innovation variables are measured at the firm level. Of studies that advocate a sand and grease relationship (*alternate argument 1*), firm-level innovation measures appear in 58 % of articles (7). Of research that confirms a negative relationship between corruption and innovation, the ratio of firm-level innovation is similar at 59 % (19 articles). Once again, the concentration of firm-level measures in studies that find a positive relationship verifies the negative externalities of corruption. Corruption produces direct and indirect costs, the latter of which include a poorly functioning institutional infrastructure, erosion of trust between business partners, and reduced foreign investment. While aggregate innovation variables can incorporate the indirect negative effects of corruption that impact all firms in a context, firm-level innovation variables capture the potential direct benefits of corruption to bribing firms. The net welfare loss is especially significant when bribing firms are in a position to secure access to government resources, such as permits and contracts, at the exclusion of the majority.

Researchers also categorize firm-level innovation output based on type: product, process, service, marketing, and organizational innovation (e.g. Riaz and Cantner, 2020). Product innovation tends to be the focus of research as opposed to the other types. This conceptualization is consistent with the Schumpeterian definition of innovation that encompasses novel firm activities as opposed to exclusively technologically focused product innovation (Ahuja et al., 2008). We find that the granularity of innovative activities measured impacts the consistency of results. In all 70 articles that conceptualize a causal relationship (Framework 1), 21 articles use self-reported, simplistic measures of innovation (e.g. a dummy variable indicating whether a firm engages in product, process, marketing, or organizational innovation over a period of time). These measures of innovation do not specify its nature and magnitude, and findings are inconsistent among these studies. On the other hand, there is a conclusive finding that corruption negatively impacts innovation when studies use quantifiable measures, such as patents (e.g. T.V. Nguyen et al., 2016; Sivak et al., 2011) and R&D intensity (e.g. Alam et al., 2019; Rocha et al., 2019). Taking both these findings into account, corruption may have a negative impact on the types of product innovation that are more significant in nature (e.g. products new to the world vs. upgrades), as they more likely entail substantial R&D efforts and thus require patent protection.

Table 2
Hypotheses and level of objectivity of corruption measures.

	Alternate argument 3 (unclear)	Dominant argument (sanding)	Alternate argument 1 (sanding and greasing)	Alternate argument 2 (greasing)
Objective measurement used	5/14 = 35.7 %	16/36 = 44.4 %[a]	7/11 = 63.6 %[b]	6/8 = 75 %
Subjective measurement used	9/14 = 64.3 %	21/36 = 58.3 %[a]	4/11 = 27.3 %[b]	2/8 = 25 %

[a] Total is larger than 100 % because two articles contain both objective and subjective measurements.
[b] One article contains both objective and subjective measurements; another article does not conduct an empirical analysis.

5.2. Theoretical underpinning of causal claims

5.2.1. Theoretical mechanisms of the sanding perspective

The theoretical perspective employed by researchers impact the causal claims they make. In 36 articles that envisage the sanding effects of corruption on innovation, economic theories (e.g. transaction cost economics and agency theory) and economics-rooted terminology (e.g. rent, expropriation, and property rights protection) appear in 27 articles. Together with 13 articles that employ various versions of the institutional perspective,[3] the sanding argument receives strong support from both economics and sociology.

Table 3 visualizes the theoretical mechanisms underlying the sanding effects of corruption. At the transactional level, the commonly used logic is that corruption increases the costs of doing business such as getting loans and licenses (e.g. Sena et al., 2018; Anokhin and Schulze, 2009). Correspondingly, fewer resources are allocated to innovative activities (Paunov, 2016). Corruption also erodes trust between transacting partners by increasing the chances of agency problems such as hold-up (Park and Xiao, 2021), thereby reducing innovation. At the firm level, two competing logics exist (Xu and Yano, 2017). The

[2] In the meta-analysis, firm-level corruption was the independent variable and encompassed the percentage of sales paid towards informal payments. The dependent variable encompassed a binary indicator for engaging in product innovation. With a total of 10 studies and 32,191 observations, we find that corruption at the firm-level is positively associated with firm-level innovation (pooled effect: 0.314, $p < 0.000$), but heterogeneity exists ($Q = 1856.609$, df = 9; 9, $p < 0.000$).

[3] The total number of articles that employ institutional or economic perspectives does not add up to 36 because 7 papers simultaneously employ two streams of literature to support their arguments. In addition, 3 articles simply review previous empirical findings and do not discuss the theoretical mechanisms of their arguments. We group North's (1990) new institutional economics together with neo-institutional theory from sociology, as they tend to co-occur in the reviewed papers. Nevertheless, we respectfully acknowledge the connections between new institutional economics and subsequent economic theories (e.g. transaction cost economics, property rights theory, and agency theory).

Table 3
Theoretical mechanisms of the causal link.

Unit of analysis	The sanding perspective			The greasing perspective		
	Logic	Freq.	Mechanisms	Logic	Freq.	Mechanisms
Transactional level	Cost or transaction cost logic	19	• High level of corruption means higher costs of transactions • Fewer resources can be allocated to innovation	Cost or transaction cost logic	16	• Corruption can lower costs of transactions if: 1. Corruption is petty, routinized, and certain. 2. Adverse effects (increased transaction costs) are unavoidable if firms choose not to bribe. • After paying bribes, residual resources can be allocated to innovation.
	Agency theory	15	• Corruption lowers trust between actors • Agency problems (e.g. hold-up) occur. • Firms are less willing to engage in innovation that requires transacting with other actors			
Firm level	Expropriation logic	7	• Corruption increases risk of expropriation by bureaucrats			
	Rent seeking logic	10	• Rent from bribery is larger than rent from innovation • Resources diverted from innovation to corruption			
				Strategic management perspective	7	• Properly orchestrated non-market strategies increase the effectiveness of a firm's market strategies. o [This perspective is often used in tandem with the transaction cost logic]
Institutional level	Institutional perspective	13	• Formal and informal institutions are supporting pillars of innovation. • Formal institutions (e.g. property rights protection) weaken with rampant corruption. • Informal institutions (e.g. culture) weaken with rampant corruption.	Institutional perspective	12	• Informal institutions can substitute formal institutions for navigating corruption when the latter is weak o [This perspective is often used in tandem with the transaction cost logic]
International level	FDI logic	2	• FDI is a key source of new technologies. • Corruption leads to lower levels of inward FDI.			

expropriation logic sees firms as the victims of corruption. Corrupt environments not only increase transaction costs, but increase the risk that returns from innovative activities are expropriated by bureaucrats (Goel and Nelson, 2018). The rent-seeking logic, on the other hand, views firms as active participants in corruption. When firms realize that rents from bribery are higher than rents from innovation, they shift from innovation activities to corrupt exchanges (Anokhin and Schulze, 2009; Athanasouli and Goujard, 2015). While associated with similar outcomes, the expropriation and rent-seeking logics have differing underlying behavioral assumptions. In addition, although the expropriation logic appears less frequently than the rent-seeking logic (7 articles as opposed to 10), the former logic receives stronger support when the two logics are examined simultaneously (Lin et al., 2021; Xu and Yano, 2017). In other words, in the event that firm-level innovative activities are hampered by corruption, it is more likely due to the exogenous pressure from corrupt bureaucrats than driven by a firm's rent-seeking decision. (See Table 4.)

The sanding effects of corruption also manifest at the macro-level. Building on North's (1990) argument on institutions and economic growth, one strand of literature suggests high levels of corruption deteriorate the formal institutional framework of a country (e.g. property rights protection and the rule of law), deterring organizations from engaging in innovative activities (Wellalage and Thrikawala, 2021). Corruption also erodes the informal institutions of a country (e.g. values, culture, and social norms) that reduce transactional uncertainty and support formal institutions (Ghura et al., 2017). Because formal institutions are essential for substantial and risky innovative efforts (DiRienzo and Das, 2015), high levels of corruption produce low levels of innovation in a country.

At the international level, some researchers identified foreign direct investment (FDI) as an intermediary that bridges the causal link between corruption and innovation. Sena et al. (2018) argues that corruption harms a country's ability to attract FDI, often the key source of new technologies and capabilities. However, researchers have not reached a consensus on the role of FDI. In Section 5.3, we explain that researchers also theorize FDI as an exogenous variable moderating the corruption-innovation relationship (e.g. Smith and Thomas, 2015).

Overall, the theoretical mechanisms supporting the sanding perspective span across levels—from the micro, transactional level to the macro, institutional level—and draw heavily on economic literature. This is in sharp contrast to the theoretical mechanisms underlying the greasing perspective.

5.2.2. *Theoretical mechanisms of the greasing perspective*

Scholars tend to employ multiple theories to argue for when the positive effects of corruption unfold. Since there are fewer articles taking this perspective, we survey 19 papers from both *alternate arguments 2* (greasing) and *1* (both sanding and greasing) to uncover causal explanations.

At the transactional level, the greasing perspective argues that corruption promotes innovation by reducing transaction costs (Barasa, 2018). Thus, it is employed in the opposite way to the sanding perspective. To the degree that bribes can speed up and reduce the uncertainty of bureaucratic processes impacted by corruption, such as permit applications, firms can focus more on value-adding activities such as innovation (e.g. Goedhuys et al., 2016; Riaz and Cantner, 2020; Xia et al., 2021). At face value, this strand of literature contradicts the transaction cost logic of the sanding perspective. However, a positive benefit to innovation occurs under two conditions: 1) an anticipated, routinized payment and 2) a limited number of options for decision makers. Regarding the first condition, the cost of a bribe should be lower than the transaction costs when a firm does not bribe (Habiyaremye and

Table 4
Moderators.

	Description/ Examples	Main arguments	Consistency of Results
Aggregate-level			
Institutional infrastructure	• Political instability: a dummy variable indicating whether a firm regards political instability as a severe problem (WBES) • Bureaucracy: the difficulty of obtaining business licensing and permits, measured on a scale of 0 (not an obstacle) to 4 (a very severe obstacle) (WBES)	• Sanding effect weakens or becomes a greasing effect in countries with worse institutional infrastructure.	Largely consistent
Economic development	• Gross domestic product (GDP) per capita (World Bank Governance Indicator)	• Sanding effects weaken in countries with higher level of economic development.	Consistent
Openness of an economy	• FDI: net inflows of foreign direct investments as a percentage of GDP of a country (Anokhin and Schulze, 2009) • Share of exports: share of exports as a percentage of gross regional product (Smith and Thomas, 2015)	• FDI strengthens the sanding effects of corruption. • FDI strengthens the greasing effects of corruption. • The greasing effect is weakened in regions with greater trade activity	Inconsistent
Natural resource rents	• Natural resource exports (Observatory of Economic Complexity)	• Rents from natural resources uphold the sanding effects of corruption.	Consistent
Industry-level			
Technological intensity	• Patent intensity of an industry obtained by dividing fractional patent count by nominal value added (Paunov, 2016) • Certificate intensity of an industry measured by the average share of establishments holding certificates (Paunov, 2016). Original data from WBES)	• Technological intensity strengthens the sanding effects of corruption	Largely consistent
Reliance on public services	• Contract dependence: share of industry inputs that are not traded on exchange markets. The assumption is that the higher level of contract dependence, the more firms in the focal industry rely on relationship-specific transactions to get intermediate inputs. Therefore, they depend more on courts, regional governments, regulatory authorities to reduce transaction costs (Athanasouli and Goujard, 2015) • A set of dummy variables that represent the identity of a firm in different industry sectors. The premise is that certain industries rely more on the public services to ensure smooth transactions (Riaz and Cantner, 2020).	• High reliance on public service strengthens the sanding effects.	Largely consistent
Firm-level			
Size and age	• Firm size: number of employees • Firm age: number of years since establishment.	• The greasing/ sanding effects of corruption are stronger in smaller firms	Inconsistent
Productivity and financial situation	• Firm productivity: the efficiency score of a firm calculated by a partial frontier analysis. Similar to data envelopment analysis (DEA), this model generates a production-possibility frontier in which the relative position of firms can be determined. • Firm financing: a set of dummies variables denoting whether a firm indicates that it needs a bank loan, applies for bank loans, and has difficulties applying for loans (Ha et al., 2022; Original data are from Vietnam SME Survey	• The sanding effects of corruption are weakened among productive firms. • The greasing effects of corruption are weakened for firms that lack financial capital.	• Inconsistent • insufficient data (only one article)

(*continued on next page*)

Table 4 (*continued*)

	Description/ Examples	Main arguments	Consistency of Results
	conducted by Central Institute for Economic Management)		
Political connections	• Political connections: top management team members' interpersonal relationships with government officials (Liu et al., 2021) • Political connections: survey questions regarding the extent to which the senior management spent time dealing with requirements of government regulations (WBES)	• Political connections strengthen the greasing effects of corruption.	Inconsistent
Governance structure (board composition, TMT gender, ownership, etc.)	• Firm ownership: whether the firm is a state-owned enterprise (Rodríguez-Pose and Zhang, 2020; Wu and Dong, 2021) • TMT gender: the gender of the highest ranking manager (WBES) • Board composition (independent boards): the proportion of independent directors in each board relative to the median proportion of independent directors in other firms in the same industry and country (Sena et al., 2018).	• The sanding effects of corruption are weakened for state-owned enterprises. • Female executives weaken the greasing effects of corruption. • Independent boards mitigate the sanding effects of corruption on innovation.	• Inconsistent • Inconsistent • Not enough data (two articles in total; one has an incomplete estimation setting.)
International presence and geographic concentration	• International presence: a subsample regression analysis of firms that internationalize (Ervits and Zmuda, 2018) • Geographic concentration: (In the U.S. context) the level of concentration of one firm's operation in the headquarter state. This variable is approximated by the number of mentions of the headquarters	• SMEs that internationalize experience stronger sanding effects. • The sanding effects are stronger for geographically concentrated firms	• Not enough data (only one article) • Not enough data (only one article)

Table 4 (*continued*)

	Description/ Examples	Main arguments	Consistency of Results
	state over the mentions of all states in a firm's 10-K report (Huang and Yuan, 2021).		
Competitive pressure from informal sector	• Informal competition: A survey question on whether practices of competitors in the informal sector an obstacle to the focal firm (WBES)	• The greasing effects of corruption are stronger when firms face competitive pressure from the informal sector	Largely consistent

Raymond, 2018; Nguyen et al., 2016a; Riaz and Cantner, 2020). If a bribe payment is uncertain and subject to increases due to further impositions imposed by bureaucrats, the costs of bribes will likely surpass any reduction in transaction costs (Blackburn and Forgues-Puccio, 2009; Tomaszewski, 2018). Regarding the second condition, in contexts where corruption is pervasive and normalized, it may be unlikely that a firm can refrain from corruption and remain afloat. In this case, firms may have no other option but to pay bribes irrespective of its costs. The second condition suggests that the greasing effect may be pertinent only in contexts with weak institutional infrastructure (e.g. Fu and Jian, 2021; Krammer, 2019; T.V. Nguyen et al., 2016), to be specified in the next section.

Researchers often complement their transaction-cost based arguments with additional theoretical lenses, the most common of which is the institutional perspective. Of the 16 articles that employ a transaction cost logic in deducing a greasing impact, 10 articles simultaneously use institutional perspectives (either new institutional economics (North, 1990) or neo-institutional theory from the sociological tradition) to narrow down the scenarios in which innovation benefits from corruption. As detailed in the previous section, both formal and informal institutions play essential roles in promoting innovation (North, 1990). Some researchers, however, argue that formal and informal institutions can be substitutive. When formal institutions are not robust, firms do not necessarily abandon their innovation efforts. Instead, they turn to informal institutions, such as building political ties, accumulating relational capital, and paying speed money, to ensure that their innovative activities proceed without obstructions (e.g. Chadee et al., 2021; Fu and Jian, 2021; Krammer, 2019). The prevalence of both corruption and innovation has been documented in countries with weak formal institutions such as China (Fu and Jian, 2021), Vietnam (N.A. Nguyen et al., 2016; T.V. Nguyen et al., 2016), and India (Jain, 2021).

In addition to institutional perspectives, management and strategy scholars also pair the transaction cost logic with non-market strategy literature to expound the greasing effect. Non-market strategy are a firm's actions "to improve its performance by managing the institutional or societal context of economic competition" (Mellahi et al., 2016). Researchers find that properly orchestrated non-market strategy may enhance the effectiveness of a firm's market strategy (Xie et al., 2019). As a non-market strategy, corruption therefore holds the potential of boosting a firm's innovation effort (Ha et al., 2022; Heo et al., 2021). Relatedly, Xia et al. (2021), T.V. Nguyen et al. (2016), and Blackburn and Forgues-Puccio (2009) elaborate the premises in which corruption (non-market strategy) elevates innovation performance. First, the briber should engage in corruption for the purpose of helping the organization rather than serving their personal interests (Xia et al., 2021). Second, bribes are paid to improve the quality of public services, especially important when public sector employees receive low salaries (Ha et al., 2022; N.A. Nguyen et al., 2016). Third, corrupt activities should be

coordinated between bureaucrats such that firms are aware of the ex-ante costs of corruption (Blackburn and Forgues-Puccio, 2009). Overall, the non-market strategy literature further specifies the conditions in which innovation can co-exist with pervasive corruption.

5.3. Contextual and firm-level factors

From our SLR, we identified country-, industry-, and firm-level moderating variables that influence the corruption-innovation relationship (Table 3).

5.3.1. Country-level variables

Multiple types of variables were identified at the country-level that help reconcile the sand and grease perspectives. Institutional infrastructure is the most prevalent type of variable. In countries with weak institutional infrastructure, the negative impact of corruption is either weakened or reverses from negative to positive. The cost of bribery activities (e.g. being caught and the penalties associated) is reduced, while the uncertainty reducing benefits of corruption are accentuated. In a cross-sectional study, Xie et al. (2019) suggest that the positive impact of corruption on innovation is stronger when firms perceive political instability to be a major problem. Similar moderating effects are demonstrated in regard to the regulatory environment (McCann and Bahl, 2017), capital markets (Dang and Yang, 2016) and informal institutions (measured as the level of societal trust) (Krammer, 2019). The only exception is a study on market segmentation in China (Ren et al., 2021). In contexts with weak institutional infrastructure, the market tends to be segmented, which limits the ability of the central government to exercise administrative power evenly across regions. Focusing specifically on innovation in the renewable energy sector, the study finds that the negative effect of corruption on the number of patents is strengthened in provinces with a high level of market segmentation.

The moderating effects of institutional infrastructure can largely explain the greasing argument (*alternate argument 2*) and partly address the argument that corruption is both sand and grease (*alternate argument 1*). Of the 8 articles that follow the greasing argument, 4 are based on China, a country with weak institutional infrastructure. Institutional variables such as environmental regulation (Fu and Jian, 2021) and obstacles to obtaining business licenses (Xia et al., 2021) determine the strength of the corruption-innovation relationship. 3 other articles include multi-country samples, but all countries are characterized by weak institutions (Krammer, 2019; Barasa, 2018; Tomaszewski, 2018). Only a single article covers a range of advanced and developing countries (Heo et al., 2021). We find that 4 of the 11 articles that follow the argument that corruption is both sand and grease employ the moderating effects of country-level institutions to explain how corruption may grease or sand the wheels of innovation.

The second type of variable is a country's level of economic development. Different from institutional infrastructure, which refers to the development of a country's governance systems, economic development refers to the economic constraints of innovation such as growth, access to capital, and availability of technology. Two studies from different perspectives are devoted to this topic but produce consistent results. DiRienzo and Das (2015) finds that the level of development weakens the beneficial effects of lower levels of corruption on innovation. They reason that further reduction of corruption does not lead to a significant increase in innovation in highly developed countries. On the other hand, Bukari and Anaman (2021) find that the negative impact of corruption on innovation is more prevalent for lower-middle-income and African countries. Thus, the sanding impact of corruption on innovation seems to be stronger in countries with lower levels of economic development.

The third type of variable is a country's openness to the global economy. Researchers focus on inward foreign direct investment (FDI) and share of exports in the economy, producing varying findings. When corruption acts as sand, less corruption should lead to more innovation. There is support for the idea that the positive impact of lower levels of corruption is more pronounced in countries or regions with high inward FDI and exporting activities (Anokhin and Schulze, 2009; Smith and Thomas, 2015). However, researchers also find that the corruption-innovation relationship becomes positive and even stronger in regions of Russia with more FDI (approximated by MNE presence) (Smith and Thomas, 2015). These findings point to two plausible effects—the corruption-reducing effects of FDI (Kwok and Tadesse, 2006) or the harmful effects of foreign firms taking advantage of corruption. The latter explanation is further evidenced by Habiyaremye and Raymond (2018), which finds that foreign competitors' involvement in corruption reduces R&D and product innovation of domestic firms. The disproportionate influence over government officials held by foreign firms produces favorable outcomes at the expense of domestic firms.

The fourth type of variable is a country's reliance on natural resources. Findings in the literature align with the idea of the "resource curse", the paradox of countries with an abundance of natural resources having low levels of economic development. A contributing factor is that the monopolistic advantages of natural resources both foster corruption and inhibit innovation. A negative impact of natural resource abundance is found in countries with authoritarian regimes, a proxy for corruption (Rosenberg and Tarasenko, 2020). Namazi and Mohammadi (2018) investigate whether innovation in national resource focused economies can motivate actors to move beyond advantages gained through corruption, further attesting to the negative relationship between corruption and innovation in natural resource-dependent economies.

5.3.2. Industry-level variables

Less attention has been paid to how industry-level variables impact the corruption-innovation relationship. Of the 70 articles that conceptualize a causal relationship, 5 articles employ a multi-industry sample and examine the effects of industry-level moderators. Across the 5 articles, corruption impairs innovation to a greater extent in industries that (1) are technologically intensive (Ha et al., 2022; Paunov, 2016) or (2) require high levels of government involvement (Athanasouli and Goujard, 2015; Riaz and Cantner, 2020). Technologically-intensive firms engage in innovation that more likely requires interacting with government officials, increasing the chances of extortion in corrupt contexts. Similarly, managers of firms that rely on public services more likely face bureaucratic backlogs created by corrupt officers that necessitate illegal payments (Athanasouli and Goujard, 2015). Huang and Yuan (2021) corroborate this logic at the firm-level by showing that corruption more severely impacts innovation for firms that sell in greater proportion to government.

Industry-level factors can partly explain the argument that corruption is both sand and grease (*alternate argument 1*). Among the 11 articles under *alternate argument 1*, 3 (Ha et al., 2022; Riaz and Cantner, 2020; T. V. Nguyen et al., 2016) suggest that the sand and grease effects hinge on industry. In studies drawing on a single industry, we observe stronger sanding effects for technologically intensive industries. Gouvea et al. (2012) find that lower levels of corruption are associated with a greater level of patenting and product introductions in the nanotechnology industry. Low corruption environments are a more likely choice for the R&D investment of firms in the renewable energy industry (Lv and Spigarelli, 2016). Clinical trials, key to the innovation process in the pharmaceutical industry, are more likely located in countries with low levels of corruption (Singh and Wang, 2013).

5.3.3. Firm-level variables

Compared to country-level variables, findings regarding the influence of firm-level variables on the corruption-innovation relationship are less consistent. We classify the various firm-level characteristics examined in the literature into six types and find some particularly useful in explaining the causal relationship between corruption and innovation observed by researchers.

Firm size and age are frequently examined moderators and produce inconsistent findings (e.g. Ha et al., 2022; Lin et al., 2010). One

perspective is that larger and older firms possess an abundance of resources that buffer them from the negative impact of corruption. The other perspective is that the innovation of larger and older firms is affected more by corruption (e.g. Riaz and Cantner, 2020). The impact of size is especially muddled when researchers simultaneously examine various innovation activities of large and small firms in response to corruption (Tomaszewski, 2018).

Firm productivity and financing variables produce confounding results. More productive and affluent firms hold dominant market positions and are at the technological frontier of their industry, which provides them an abundance of resources and strategic flexibility. As a result, they can avoid business exchanges that are susceptible to corruption (Namazi and Mohammadi, 2018). Corruption thus has a weaker impact on the innovation of more productive firms. This idea is confirmed in a quantile regression (Rocha et al., 2019) but cannot be replicated in sub-sample analyses (Lesáková et al., 2017). The inconclusive role of firm productivity partly explains arguments with no prediction (*alternate argument 3*); in the fourteen articles that do not propose a clear relationship between corruption and innovation, two examine the moderating effects of firm productivity (Lesáková et al., 2017; Rocha et al., 2019). The need for external financing places limits on the greasing impact of corruption; Ha et al. (2022) finds that the positive impact of corruption on innovation is weaker for firms that either applied or indicated the desire to apply for a loan.

Political connectedness variables also produce opposing moderating results. A firm is considered politically connected when its top-level managers have interpersonal ties with elected representatives or government officials. One line of research indicates that connections attenuate the relationship between corruption and both innovation input (Yu et al., 2019) and output (Krammer and Jimenez, 2020; Liu et al., 2021). These studies suggest that political connections serve as substitutes for corruption as a means of increasing innovation. However, Dang and Yang (2016) find that political connections become conduits for engaging in corruption in a sample of listed companies in China, which points to the complementary role of political connections in the corruption-innovation relationship.

The results from governance-related characteristics are the most contradictory of all firm-level moderators. Three major governance variables are examined: ownership structure, top management team gender, and board characteristics. Among the 4 papers dedicated to the moderating effects of ownership structure, 2 articles find that the negative effect of corruption on innovation (sanding effect) is weaker for state-owned enterprises (Gan and Xu, 2019; Rodríguez-Pose and Zhang, 2020). 1 article found the opposite result (Lin et al., 2021) while another found an insignificant result (Xu and Yano, 2017). Regarding senior management's gender, Wellalage and Thrikawala (2021) and Xia et al. (2018) produce contradictory findings regarding the likelihood that female executives leverage a corrupt environment to increase firm innovation. Regarding board characteristics, Sena et al. (2018) show that independent boards mitigate the negative impact of corruption on firm innovation. However, the weakening effects of independent boards is yet to be replicated in other research.

The fifth type of firm-level variable is a firm's international presence and geographic concentration. Regarding international presence, corruption has a stronger negative impact on the innovation of SMEs that internationalize. High levels of country-level corruption decrease rates of patenting for innovation-focused SMEs that sell globally (Ervits and Zmuda, 2018). In regard to geographic concentration, Huang and Yuan (2021) find that the innovation activities of more geographically concentrated US firms are more greatly impacted by corruption. Geographic concentration leaves firms susceptible to corruption by creating greater dependency with state governments. With only one article on each topic, the reliability of findings regarding international presence and geographic concentration cannot be determined.

The last type of firm-level variable is competitive pressure from the informal sector. Informal competitors are firms that are not legally registered and thus not beholden to the same formal requirements as registered firms. Studies that consider the informal sector can help make sense of the greasing argument (*alternate argument 2*). Ha et al. (2022) find that compared to informal firms, formally registered firms experience a stronger greasing effect of corruption.

6. Discussion

After in-depth analysiss of the antecedents, consequences, moderators, and theoretical mechanisms of the corruption-innovation relationship, we can now uncover general patterns of contingencies that determine the sanding or greasing impact of corruption on innovation. We visualize the general patterns in Fig. 4 and discuss them further in Section 6.1. Thereafter, we discuss theoretical and policymaking implications our of study, as well as empirical suggestions for future research.

6.1. Patterns underlying the corruption-innovation relationship

6.1.1. The sanding pattern

"Pattern" in our paper refers to the social-economic alignments in which a proposed theoretical relationship most likely holds (Heiberg et al., 2022). In this study, patterns consist of the measurements and levels of analysis of corruption and innovation, the relevant contextual and firm-level factors impacting the relationship, and dominant theories used in the relationship building process. Fig. 4 lists all relevant factors and displays the sanding and greasing patterns. We use the terms "most likely" and "likely", roughly defined by the frequency of counterexamples in the literature database, to qualitatively describe the likelihood of a factor being associated to each pattern.

The sanding pattern typically includes aggregate measures of corruption and innovation. When corruption is conceptualized at the region or country level, it captures both the micro-level impacts to a single firm and the impacts to firms more broadly (e.g. market distortions and efficiency loss). When innovation is conceptualized at the state or country level, it encompasses the interconnected nature of innovation in an ecosystem (via spillover or agglomeration, see Ning et al., 2016). Thus, at aggregate levels of analysis, the sanding pattern captures how corrupt contexts reduce overall levels of innovation. To a lesser degree, the sanding pattern can also be observed at the firm level. Firms suffer from corruption when its benefits do not accrue in the form of innovation outcomes or when corruption is measured in terms of competitors' bribing activities. The sanding impact of corruption most likely applies to innovation efforts subject to high levels of R&D, the outcome of which is often a patented product. Regarding the country-level moderators, the sanding impact is prevalent in countries that have lower levels of economic development and a high reliance on natural resources. At the industry level, the sanding pattern is more pronounced in industries that are technologically-intensive and rely on public services. At the firm level, public firms that lack board independence, internationally-focused SMEs, and geographically concentrated firms are associated with a stronger sanding effect. In terms of the theoretical mechanisms of the sanding pattern, researchers draw upon economic theories (e.g. transaction cost economics and agency theory) to develop their arguments. Increased transaction costs and decreased trust between transacting partners are the key theoretical mechanisms that establish a connection between high levels of corruption and low levels of innovation.

6.1.2. The greasing pattern

The greasing pattern is strongly associated with firm-level measures of corruption and innovation. When innovation is conceptualized at the firm-level, it has the potential to capture the potential benefits of bribery that outweigh its costs for some firms. Regarding contextual factors, the greasing pattern is more pronounced in countries with weak institutional infrastructure. At the firm level, the only consistent pattern we

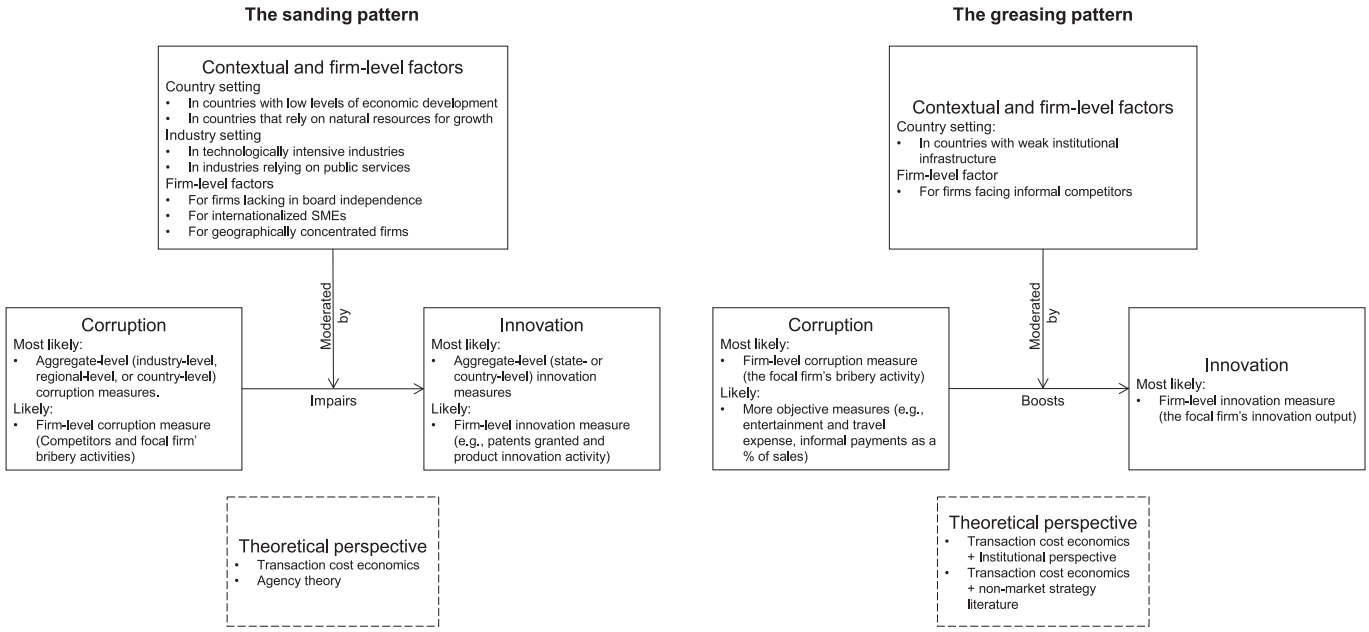

Fig. 4. Two major patterns of causality between corruption and innovation.
Note: "Most likely" means the measurement is very likely to appear in the described pattern because there is unanimous support from the results of the SLR; "Likely" means that the measurement is likely to appear in the described pattern because there are largely consistent results in the SLR (i.e. no more than one counterexample).

find is a stronger greasing effect when firms compete in an industry with a strong informal sector. Researchers propose that the greasing pattern is associated with other firm-level moderators, such as a firm's productivity (Rocha et al., 2019) and political connections (Yu et al., 2019), but the results are generally inconsistent. In the next section, we discuss future research directions for exploring how firm-level characteristics relate to a firm's ability to innovate in corrupt contexts.

Regarding theoretical mechanisms, researchers tend to combine economic theory (transaction cost economics) with institutional perspectives or non-market strategy literature to argue for a greasing effect. The common approach is to explain conditions in which the net benefit of engaging in corruption (i.e. innovation returns deducted by bribery costs) is larger than the opportunity costs of not engaging in corruption (i.e. innovation returns deducted by increased transaction costs). We find that three general conditions are associated with a greasing pattern. First, bribes are petty in nature and available widely to all firms, such that corruption can broadly increase innovation in an economy. Second, corruption is exclusionary such that if affords advantages for some firms, increasing more significantly the innovation of relatively few firms at the expense of the broader economy. Third, firms have no choice but to engage in corruption since the essential actions required for innovation require bribe payments. Institutional perspectives and non-market strategy literature are employed to establish the context specificities of the greasing pattern.

6.2. Theoretical implications

The discussion of corruption's impact on innovation should be reframed in light of findings established in this paper, as summarized by the greasing and sanding patterns. Rather than posing the simplistic question of whether corruption acts as sand or grease, a starting point for posing more nuanced questions is to acknowledge the definitive conditions under which each effect is most likely to occur.

Whether conditions exist that allow for exclusionary benefits can serve as a theoretical starting point in predicting and developing arguments for a greasing impact. The motivation underlying corruption can be telling of whether a firm stands to gain exclusionary benefits. Extant literature tends to assume corruption as an exogenous problem imposed on firms. However, some firms actively seek out bribing opportunities to gain exclusive advantages in the market (Cuervo-Cazurra, 2016). A sand or grease effect may therefore hinge on whether firms "endogenize" corrupt activities as part of their strategy or if it is an imposed payment.

Whether a firm's environment affords some firms exclusionary benefits can also be used as a basis for developing theoretical arguments. The distinction between incomplete and captured institutions is relevant because it is telling of the scope of actors that can benefit from corruption. In incomplete institutions, bribery acts as a form of "speed money" available to all firms to reduce uncertainty in business activities, including innovation (Habiyaremye and Raymond, 2018; N.A. Nguyen et al., 2016). In captured institutions, engaging in corruption is only possible for certain firms that the state prioritizes and affords privileges to (Acemoglu and Robinson, 2013; Dorobantu et al., 2017: 125). While a greasing impact may exist in each context, the benefits to corruption for privileged firms should be greater in captured institutions. Thus, future research should pay closer attention to whether a country or industry environment is subject to regulatory capture.

Future research can also focus on the interdependency between the greasing and sanding impacts to understand more precisely the impact of corruption across firms in a given context. For example, Rosenberg and Tarasenko (2020) finds a sanding effect at the country level for natural resource reliant economies, a context in which the scarcity of resources and government control enables the provision of exclusionary benefits for some firms. As a counter-point, future research can aim to understand which firms are most likely to obtain exclusionary benefits in natural-resource reliant economies, whether the level of corruption they experience is positively associated with their innovation output, and whether the directionality of effect differs relative to those firms unable to obtain these benefits.

The inconsistent moderators identified in this paper can be used by future researchers to further identify the unknown theoretical mechanisms driving various conceptions of the corruption-innovation relationship. According to our SLR, inconsistent findings are concentrated in studies employing firm-level moderators. The role of political connections in particular should be clarified, given its complicated interactions with corruption in shaping a firm's innovation input and performance. Do they complement or substitute each other in innovation activities?

Researchers find that the resource-acquisition benefits of political connections may be heightened when corruption allows for exclusionary benefits for some firms, increasing innovation (Barber and Diestre, 2019). On the other hand, the resistance to new sources of information (Horak et al., 2020) caused by political ties may drive the overall effect when corruption does not provide any privileged access, reducing innovation. Future research can consider the nature and measurement of innovation (e.g. R&D intensiveness, patents), corruption (e.g. level of analysis), and the context (e.g. degree of government involvement in an industry, extent of regulatory capture) in theorizing whether political connections will interact with corruption to improve or hinder innovation.

Future research can also examine firm-level moderators not yet considered in the extant literature with the findings of this SLR in mind. Given their association with illegal conduct, surprisingly few studies examined accounting and financing-related practices. Their further examination can shed light on the types of firms that can take advantage of corruption to improve their innovation outcomes. For example, the level of reporting transparency (based on legal status or regulatory requirements) may have an influence by impacting the degree to which costs incurred for engaging in corruption can remain hidden. Greater transparency may result in an inability to hide significant bribery activity, which may have a greater long-term benefit for the average R&D intensive firms focused on developing patentable new products. On the other hand, it may harm those firms in the best position to obtain exclusive resources from government (e.g. state-owned enterprises, large and resourceful multinationals).

6.3. Policy implications

Our findings paint a clear picture that while conditions exist in which corruption fosters innovation, the long-term impact of corruption on broader innovation in an economy will be unequivocally negative. Continual innovation at the national level requires a broader ecosystem of innovative firms that support one another through their inputs and efforts, including competitors, suppliers, and distributors (Oh et al., 2016). The benefitting of privileged actors that obtain exclusive resources in corrupt environments comes at the detriment of most firms. Thus, policies lax on corruption due its greasing impact will benefit relatively few firms at the expense of a strong innovation ecosystem.

Our findings make clear that corruption tends to produce the strongest negative impact on R&D and patents, the types of innovation activities argued to be most conducive for economic growth (McCarthy et al., 1987). In addition, corruption may further incline developing country firms towards process innovations and re-enforce their position in global value chains (Mohammad and Husted, 2021). These findings affirm the question of whether privileged firms in corrupt environments are truly innovative, since monopolistic advantages can foster complacency by suppressing competition (Wegner, 2019). Innovation in technologically intensive industries, including the environmental sector, is also suppressed by corruption (Fu and Jian, 2021; Ren et al., 2021). Advancements in the environmental sector can boost economic productivity as well as improve a country's social welfare, making this finding especially problematic.

Also troubling are the findings that foreign firms in developing countries are more likely to experience a greasing impact of corruption whereas corruption reduces the innovation of internationalized domestic firms. Thus, the greasing impact of corruption favors international firms at the expense of domestic firms with aspirations to compete globally, causing the returns to innovation to be concentrated outside of countries in which the greasing effect occurs.

The implications of the greasing impact of corruption on innovation are made even more apparent when considering the differing findings pertaining to institutional infrastructure and economic development. A weak institutional infrastructure, defined in terms of quality of governance, is most often associated with a lower level of economic development. A weak institutional infrastructure can give rise to a greasing effect, whereas low levels of economic development are associated with a sanding impact to innovation. The reason for these different effects is because weak institutional infrastructure is associated with poorly functioning governance systems, which create uncertainty that is reduced by corrupt exchanges. On the other hand, economic constraints to innovation, such as the availability of technology, level of education, and foreign investment inflows, are further worsened by corruption (Mo, 2001). While corruption might increase innovation in contexts with weak institutional infrastructure, it is unlikely to translate into the type of innovation output required for long-term economic growth. Corruption results in the suppression of economic factors required for high-growth innovation efforts such as R&D, causing growth to stagnate. For instance, weaker institutional infrastructure, higher levels of corruption, and a lack of technological innovation underlies a slower growth trajectory that causes countries in Central and Eastern Europe to remain in the "middle-income trap" (Shkolnykova et al., 2024).

Overall, corruption's greasing effect can only be considered a "second-best" option under conditions of weak institutional quality. Furthermore, countries where corruption facilitates exchanges broadly across the economy will still feel the brunt of its negative societal implications, such as increasing levels of poverty and inequality (Kaufmann, 1997). Therefore, policymakers should abstain from a shorter-term perspective in which corruption is allowed for the purpose of serving as speed-money or fostering the innovation of privileged firms. Moreover, lower levels of corruption are conducive for meeting increasingly important policy objectives (Kaplinsky and Kraemer-Mbula, 2022; Audretsch et al., 2020). Based on corruption's clearly negative impact on innovation in the high-tech sector, anti-corruption policy in the developing country context will increase the capability of firms to leapfrog into nascent industries such as information and communication technology (ICT), continually innovate, and become internationally competitive. Since the sanding effect applies to firms with a high reliance on government and a focus on developing patents, innovative start-ups, having both these characteristics, should also flourish in low corruption environments. Anti-corruption initiatives should also disproportionately impact international firms operating in a country, having an overall benefit to domestic companies.

6.4. Empirical challenges and suggestions for future research

We outline key empirical issues and make corresponding suggestions for future research. We delineate our discussion based on the three stages of empirical research: data collection, methodology, and presentation of results.

6.4.1. Data collection

6.4.1.1. Self-responses. Firm-level measures of corruption employed in the literature are primarily self-responses, which are subject to social desirability bias (Svensson, 2003). Exacerbating the issue is that surveys are often administered with the help of governments (e.g. WBES), creating fear of reprisal (Jensen et al., 2010). Recent research has started to use objective firm-level corruption data such as expenditures on entertainment to mitigate bias (e.g. Xu et al., 2019). Another approach is to document and control nonresponse in regression models based on measures of national political freedom (Jensen et al., 2010).

6.4.1.2. Ambiguous measurements. Ambiguous measures lower the validity and reliability of findings. For example, many studies measure product innovation using a simplistic, self-reported dummy variable. To tighten the connection between theory and empirics, future research should phase out such measures and incorporate those that better capture the nature and significance of innovation. For example, researchers

may differentiate radical, incremental, modular, and architectural innovation in their research design (Henderson and Clark, 1990). Incorporating innovation inputs as a potential confounding factor may also lead to greater consistency in results. In addition, researchers can employ output measures that have received insufficient attention (see Appendix F for a review of innovation measures). One example is trademarks. As an intangible asset, trademarks indicate the branding and service innovation of a company (Flikkema et al., 2019). Future research could use trademarks to delve into the heterogenous impact of corruption on the types of innovation they are most associated with (i.e. marketing and service innovation) to derive more pertinent recommendations for policymakers.

6.4.1.3. Alternative datasets. Although WBES has been widely used in the extant literature, its coarse measures,self-response bias, and sample anonymity has prevented researchers from in-depth analyses of the theoretical mechanisms linking corruption to innovation. We recommend that researchers compile original datasets by merging public databases (e.g. Compustat) with other patent and corruption data. An underutilized dataset is violations of the Foreign Corrupt Practices Act (FCPA) which covers corruption cases in the United States (Burnham et al., 2018). However, researchers should account for selection bias since only firms caught for engaging in corruption are included in FCPA data. Researchers investigating the corruption-innovation link in developing countries can use the Chinese Patent Data Project (CPDP). The CPDP links patents registered in China to all listed firms and their subsidiaries (He, Tong, Zhang, and He, 2017). This dataset has rich empirical potential since it can be combined with various corruption variables and controls for confounding variables.

6.4.2. Methodology

6.4.2.1. Endogeneity. Omitted-variable bias and reverse causality, the two major causes of endogeneity, contribute to discordant results in the extant literature. Omitted-variable bias is especially pronounced in aggregate-level analyses since it is difficult to control all confounding factors. One solution is to employ a panel data model with fixed effects (e.g. Forson et al., 2021; Jain, 2021; Ren et al., 2021; Wen et al., 2020). Using a fixed-effects model, researchers can remove the disturbance to the relationship from unobserved, unchanged country-level factors. Reverse causality is a relevant issue at the firm-level. As our conceptual mapping revealed, a firm's innovation activities can impact the extent to which it faces corruption (the *reversed causal framework* in Fig. 3, see also Ayyagari et al., 2011; Burnham et al., 2018). One solution is to utilize a valid instrumental variable (e.g. Krammer, 2019; Goel and Saunoris, 2020). Alternatively, researchers can identify an exogenous shock strongly related with the level of corruption in the context and reasonably uncorrelated with innovation. One example is China's anti-corruption campaign initiated in late 2012 (Xu and Yano, 2017; Giannetti et al., 2021). The campaign was not directly intended to impact innovation, making it exogenous to the corruption-innovation relationship. Several articles have utilized this exogenous event (e.g. Gan and Xu, 2019; Park and Xiao, 2021; Xu and Yano, 2017).

6.4.2.2. Qualitative research. Given the difficulty of eliminating the endogeneity concern, researchers should consider qualitative approaches to capture insights pertaining to the directionality of the relationship as well as unexplored motivations for engaging in or refraining from corruption. Our SLR suggests that there is limited qualitative research on the topic (e.g. Ghura et al., 2017; Gong and Yang, 2020). That being said, researchers should be aware of the heightened risk of social desireability bias when conducting interviews regarding sensitive topics such as corruption.

6.4.3. Presentation of results

6.4.3.1. Marginal effects. A common approach for evaluating whether corruption acts as a sand or grease is to incorporate a moderator into the model. To visualize the results, researchers tend to use a moderation effect diagram that shows how the slope of a fitted regression changes at high and low levels of the moderator. This method is limited since it only visualizes the difference in slope at two levels without detecting the inflection point at which a change in directionality occurs (i.e. negative to positive or vice-versa). Detecting an inflection point is essential for theoretical arguments that put forth conditions that determine whether a sanding or greasing impact exists. We recommend that researchers use a marginal effects figure instead (i.e. partial derivative of y with respect to x) that specifies the effect of corruption at every level of the moderator.

7. Concluding remarks

Why is corruption theorized as harming innovation in some studies and benefiting innovation in others? We attempted to solve this puzzle through a SLR of articles published between 2004 and 2021. We first identified four types of conceptualizations between corruption and innovation: causal, substitutive, symbiotic, and reversed causal. Then, we conducted an in-depth antecedent-consequence-moderator analysis for the studies that make a causal claim between corruption and innovation. We found that whether corruption sands or greases innovation hinges on three factors: (1) the measurement and conceptualization of corruption and innovation; (2) the theoretical perspective employed; and (3) the contextual and firm-level factors that influence the corruption-innovation relationship. Our SLR and conceptual development is much needed considering the literature's variability regarding theoretical approaches, measures, contexts, and levels of analysis. It is our hope that the findings of this study be used as a starting point for future research to shift focus beyond the dichotomy of corruption as sand or grease. By being cognizant of the core conceptual, operational, and theoretical determinants of whether corruption decreases or increases innovation, researchers can pose more nuanced questions of strategic and policy-making relevance.

CRediT authorship contribution statement

Shoeb Mohammad: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jie Yang:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – review & editing. **Irfan Butt:** Data curation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Coding scheme

Coding scheme: research design

Category	Coding
Study type	Conceptual, empirical
Data type	Qualitative, quantitative, no data
Data source	Primary, secondary
Number of countries in sample	Actual
Country of data collection	Actual

Coding scheme: predicted relationship

Category	Coding
Hypothesis proposed	Actual
Direction of hypothesis	Positive, negative

Coding scheme: variables

Category	Coding
Variable	Actual
Variable classification	Antecedent, mediator, moderator, control
Variable level	Country, region, industry, firm
Variable operationalization	Actual
Corruption variable operationalization type	Subjective, objective
Innovation variable operationalization type	Input, output

Appendix B. Antecedent-consequence-moderator analysis illustration

Study ID	Actual variable from study	Variable consolidation	Higher categorization	Operationalization type
37	R&D investment	R&D	Innovation	Outcome
78	R&D elasticity	R&D	Innovation	Outcome
89	Enterprise R&D investment	R&D	Innovation	Outcome
495	Firm-level R&D	R&D	Innovation	Outcome

Appendix C. List of studies

ID	Title	Journal	Year
12	Revisiting the relationship between corruption and innovation in developing and emerging economies	Crime Law and Social Change	2019
18	Entrepreneurship in autocratic regimes - how neo-patrimonialism constrains innovation	Journal of Evolutionary Economics	2019
20	Corruption and New Product Innovation: Examining Firms' Ethical Dilemmas in Transition Economies	Journal of Business Ethics	2019
22	Government efficiency and enterprise innovation? evidence from China	Asian Journal of Technology Innovation	2019
34	The evaluation of competitive position of EU-28 economies with using global multi-criteria indices	Equilibrium-Quarterly Journal of Economics and Economic Policy	2019
37	Does anti-corruption campaign promote corporate R&D investment? Evidence from China	Finance Research Letters	2019
41	How does bribery affect a firm's future growth? Empirical evidence from transition economies	Post-Communist Economies	2019
59	Greasing the Wheels of Change: Bribery, Institutions, and New Product Introductions in Emerging Markets	Journal of Management	2019
67	Firm Development and Bribery: An Empirical Study from Latin America	Atlantic Economic Journal	2019
78	Corruption, bureaucracy and other institutional failures: the "cancer" of innovation and development	Economics Bulletin	2019
81	A cointegration analysis of tax evasion, corruption and entrepreneurship in OECD countries	Economic Research-Ekonomska Istrazivanja	2019
83	DOES CORRUPTION AFFECTS BUSINESS INNOVATION? INSIGHTS FROM EMERGING COUNTRIES	Journal of Business Economics and Management	2019
84	Does corruption slow down innovation? Evidence from a cointegrated panel of US states	European Journal of Political Economy	2019
89	REGIONAL ANTI-CORRUPTION EFFORT, POLITICAL CONNECTIONS AND FIRM INNOVATION EFFORT: EVIDENCE FROM CHINA	Bulletin of Economic Research	2019
91	Institutional determinants of R&D investment: Evidence from emerging markets	Technological Forecasting and Social Change	2019
95	Natural resource dependence and economic growth: A TOPSIS/DEA analysis of innovation efficiency	Resources Policy	2018
98	A cross-country comparison of the effects of institutions on internationally oriented innovation	Journal of International Entrepreneurship	2018
99	CORRUPTION AND GROWTH: THE PRODUCTIVITY GROWTH NEXUS	Singapore Economic Review	2018
103	Corruption, transaction costs, and innovation in Africa	African Journal of Science Technology Innovation & Development	2018
109	Corruption and Technological Innovation in Private Small-Medium Scale Companies: Does Female Top Management Play a Role?	Sustainability	2018

(*continued on next page*)

(*continued*)

ID	Title	Journal	Year
110	Determinants of process innovation introductions: Evidence from 115 developing countries	Managerial and Decision Economics	2018
112	Board independence, corruption and innovation. Some evidence on UK subsidiaries	Journal of Corporate Finance	2018
113	CORRUPTION - A DARK SIDE OF ENTREPRENEURSHIP. CORRUPTION AND INNOVATIONS	Prague Economic Papers	2018
123	Does quality of innovation, culture and governance drive FDI?: Evidence from emerging markets	Emerging Markets Review	2018
135	How do foreign firms' corruption practices affect innovation performance in host countries? Industry-level evidence from transition economies	Innovation-Management Policy & Practice	2018
141	Public Procurement in Russia: What Hinders Innovation?	International Journal of Public Administration	2018
142	Institutions, human capital and entrepreneurial orientation: implications for growth policy	Journal of Entrepreneurship and Public Policy	2018
148	Innovation Leaders, Modest Innovators and Non-innovative SMEs In Slovakia: Key Factors and Barriers of Innovation Activity	Organizacija	2017
163	How does anti-corruption affect corporate innovation? Evidence from recent anti-corruption efforts in China	Journal of Comparative Economics	2017
166	The influence of competition from informal firms on new product development	Strategic Management Journal	2017
181	The governance environment and innovative SMEs	Small Business Economics	2017
192	DYNAMICS FACTORS OF UKRAINIAN ECONOMY INNOVATIVE TRANSFORMATION: MAIN POINTS AND SPECIFICS OF INFLUENCE	Marketing and Management of Innovations	2017
197	Moderating relationship of institutions for opportunity entrepreneurship and economic development Literature review and proposed conceptual framework	World Journal of Entrepreneurship Management and Sustainable Development	2017
200	A socio-political approach to exploring the innovation culture in post-socialist countries: the case of Croatia	Post-Communist Economies	2017
211	Corruption, innovation and firm growth: firm-level evidence from Egypt and Tunisia	Eurasian Business Review	2016
226	The impact of petty corruption on firm innovation in Vietnam	Crime Law and Social Change	2016
227	Strategic and transactional costs of corruption: perspectives from Vietnamese firms	Crime Law and Social Change	2016
238	The general theory of culture, entrepreneurship, innovation, and quality-of-life: Comparing nurturing versus thwarting enterprise start-ups in BRIC, Denmark, Germany, and the United States	Industrial Marketing Management	2016
241	Anti-corruption, marketisation and firm behaviours: evidence from firm innovation in China	Economic and Political Studies-Eps	2016
251	Entrepreneurial Orientation and Corruption	Journal of Business Ethics	2016
278	Quality of government and innovative performance in the regions of Europe	Journal of Economic Geography	2015
282	The Role of Foreign Direct Investment and State Capture in Shaping Innovation Outcome in Russia	Europe-Asia Studies	2015
286	Innovation and role of corruption and diversity: A cross-country study	International Journal of Cross Cultural Management	2015
288	DETERMINANTS OF INNOVATIVE ACTIVITIES: EVIDENCE FROM EUROPE AND CENTRAL ASIA REGION	Singapore Economic Review	2015
307	Innovation in State-Owned Enterprises: Reconsidering the Conventional Wisdom	Journal of Economic Issues	2014
312	Bribe Payments and Innovation in Developing Countries: Are Innovating Firms Disproportionately Affected?	Journal of Financial and Quantitative Analysis	2014
322	Clinical trials in "emerging markets": Regulatory considerations and other factors	Contemporary Clinical Trials	2013
328	Different paths, one goal: Southeast Asia as knowledge society	Asia Pacific Journal of Education	2013
330	Global patterns of renewable energy innovation, 1990–2009	Energy For Sustainable Development	2013
340	Emerging Technologies and Ethics: A Race-to-the-Bottom or the Top?	Journal of Business Ethics	2012
342	New Products and Corruption: Evidence from Indian Firms	Developing Economies	2012
361	The impact of governance and infrastructure on innovation	Post-Communist Economies	2011
368	Property rights protection and corporate R&D: Evidence from China	Journal of Development Economics	2010
386	Entrepreneurship, innovation, and corruption	Journal of Business Venturing	2009
399	Innovation or rent-seeking: The entrepreneurial behavior during China's economic transformation	China & World Economy	2008
410	Interrelationship among institutional infrastructure, technological innovation and growth. An empirical evidence	Applied Economics	2007
462	Building an effective knowledge management system in Saudi Arabia using the principles of good governance	Resources Policy	2019
470	Country-level corruption and accounting choice: Research & development capitalization under IFRS	British Accounting Review	2019
475	Deviant Versus Aspirational Risk Taking: The Effects of Performance Feedback on Bribery Expenditure and R&D Intensity	Academy of Management Journal	2019
495	Host country institutions and firm-level R&D influences: An analysis of European Union FDI in China	Research In International Business and Finance	2019
516	ILLEGAL CORPORATE BRIBERY: THE PRESSURE TO "MAKE DO" AND ACHIEVE GOALS	Journal of Marketing Theory and Practice	2018
549	The determinants of location choice Chinese foreign direct investments in the European renewable energy sector	International Journal of Emerging Markets	2016
550	Corruption's asymmetric impacts on firm innovation	Journal of Development Economics	2016
552	Corruption and management practices: Firm level evidence	Journal of Comparative Economics	2015
611	Why is corruption less harmful in some countries than in others?	Journal of Economic Behavior & Organization	2009
628	Determinants of cross-border intellectual property rights enforcement: The role of trade sanctions	Southern Economic Journal	2004
1001	Effects of Bribery on Firms' Environmental Innovation Adoption in Vietnam: Mediating Roles of Firms' Bargaining Power and Credit and Institutional Constraints	Ecological Economics	2021
1002	Design versus utility innovation: Is corruption sanding or greasing the wheels of innovation?	Managerial and Decision Economics	2020
1005	China's anti-corruption campaign and firm productivity: Evidence from a quasi-natural experiment	China Economic Review	2020
1007	Corruption, Bribery and Innovation in CEE: Where is the Link?	Journal of Business Ethics	2021
1012	Does bribery sand or grease the wheels of firm level innovation: evidence from Latin American countries	Journal of Evolutionary Economics	2021
1019	Government corruption, market segmentation and renewable energy technology innovation: Evidence from China	Journal of Environmental Management	2021
1028	China's Anticorruption Campaign and Entrepreneurship	Journal of Law & Economics	2021
1032	The Effect of Bribery on Firm Innovation: An Analysis of Small and Medium Firms in Vietnam	Journal of Asian Finance, Economics and Business	2020
1038	Corruption pays off: How environmental regulations promote corporate innovation in a developing country	Ecological Economics	2021
1046	Do political connections matter for firm innovation? Evidence from emerging markets in Central Asia and Eastern Europe	Technological Forecasting and Social Change	2020
1051	Innovation for despots? How dictators and democratic leaders differ in stifling innovation and misusing natural resources across 114 countries	Energy Research & Social Science	2020
1058	Corruption and Corporate Innovation	Journal of Financial and Quantitative Analysis	2020
1063	Does corruption grease or sand the wheels of investment or innovation? Different effects in advanced and emerging economies	Applied Economics	2021
1067	Effect of corruption on firm level innovation: Evidence from Pakistan	Economics and Business Letters	2020
1068	Corruption and innovation in private firms: Does gender matter?	International Review of Financial Analysis	2020

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ID	Title	Journal	Year
1077	China's anti-corruption shock and resource reallocation in the energy industry	Energy Economics	2021
1079	Corruption and firm innovation: a grease or sand in the wheels of commerce? Evidence from lower-middle and upper-middle income economies	Eurasian Business Review	2021
1083	Does Political Corruption Impede Firm Innovation? Evidence from the United States	Journal of Financial and Quantitative Analysis	2021
1092	Corruption, the Shadow Economy and Innovation in Spanish Regions	Panoeconomicus	2020
1094	Country governance, corruption, and the likelihood of firms' innovation	Economic Modelling	2020
1112	CORRUPTION AND INNOVATION: LINEAR AND NONLINEAR INVESTIGATIONS OF OECD COUNTRIES	Singapore Economic Review	2020
1113	Does corruption impede innovation in developing economies? Insights from Pakistan: a call for policies reforms	Crime Law and Social Change	2021
1153	Why Do Innovations Succeed or Fail? Local Anti-Corruption Reform in China	China Review-An Interdisciplinary Journal On Greater China	2020
1154	Doing good by combating bad in the digital world: Institutional pressures, anti-corruption practices, and competitive implications of MNE foreign subsidiaries	Journal of Business Research	2021
1159	Is R&D spending influenced by disinvestment and local political corruption? The case of Indian central public sector enterprises	Economic Systems	2021
1181	The cost of weak institutions for innovation in China	Technological Forecasting and Social Change	2020
1182	Varieties in state capitalism and corporate innovation: Evidence from an emerging economy	Journal of Corporate Finance	2021
1197	How governance structure, terrorism, and internationalisation affect innovation? Evidence from Pakistan	Technology Analysis & Strategic Management	2021
1208	Determinants of Sustainable Open Innovations-A Firm-Level Capacity Analysis	Sustainability	2021
1216	Independent director network and corporate innovation: evidence from a natural experiment in China	Applied Economics Letters	2021
1219	Formal and Informal Institutions: The Independent and Joint Impacts on Firm Innovation	Management and Organization Review	2021
1231	Exploring the effects of unethical pro-organizational behaviour on innovation strategy: an examination based on firm survey in China	Asia Pacific Business Review	2021
1233	Innovation quantity or quality? The role of political connections	Emerging Markets Review	2021
1292	What drives new product innovation in China? An integrative strategy tripod approach	Thunderbird International Business Review	2020
1308	National-level innovation in Africa	Research Policy	2020
1381	Political connections and green technology innovations under an environmental regulation	Journal of Cleaner Production	2021
1401	Innovation, institutions and economic growth in sub-Saharan Africa - an IV estimation of a panel threshold model	Journal of Economic and Administrative Sciences	2021

Appendix D. Descriptive characteristics of studies

	2004–2016		2017–2021		Total	
	N	%	N	%	N	%
Study type						
Conceptual	4	4 %	7	7 %	11	11 %
Empirical	23	22 %	69	67 %	92	89 %
Total	27	26 %	76	74 %	103	100 %
Data type						
Quantitative	23	22 %	68	66 %	91	88 %
Qualitative	0	0 %	1	1 %	1	1 %
No data	4	4 %	7	7 %	11	11 %
Total	27	26 %	76	74 %	103	100 %
Data source						
Primary	1	1 %	2	2 %	3	3 %
Secondary	22	21 %	67	65 %	89	86 %
Total	23	22 %	69	67 %	92	89 %
Number of countries in sample						
1	8	8 %	38	37 %	46	45 %
2–10	3	3 %	2	2 %	5	5 %
11–20	2	2 %	6	6 %	8	8 %
21–30	3	3 %	10	10 %	13	13 %
31–60	3	3 %	5	5 %	8	8 %
61–90	3	3 %	2	2 %	5	5 %
91–120	1	1 %	4	4 %	5	5 %
More than 120		0 %	2	2 %	2	2 %
Total	23	22 %	69	67 %	92	89 %
Single country sample frequency						
China	3	3 %	23	22 %	26	25 %
USA		0 %	5	5 %	5	5 %
Vietnam	2	2 %	2	2 %	4	4 %
Pakistan		0 %	3	3 %	3	3 %
India	1	1 %	1	1 %	2	2 %
Spain		0 %	1	1 %	1	1 %
Slovakia		0 %	1	1 %	1	1 %
Germany	1	1 %		0 %	1	1 %
Morocco		0 %	1	1 %	1	1 %
Russia	1	1 %		0 %	1	1 %
Nigeria		0 %	1	1 %	1	1 %
Total	8	8 %	38	37 %	46	45 %

Appendix E. Corruption measurements

	Aggregate-level (country, region, or industry)			Firm-level		
	Sources	ID	Description/example	Sources	ID	Description/example
Subjective	• Control of corruption index (CC) • Corruption perceptions index (CPI) • International country risk guide (ICRG) • World Happiness Report (United Nationals Sustainable Development Solutions Network) • Firm-level survey questions aggregated to the • region/industry/ country level	78, 91, 98, 142, 166, 227, 278, 286, 330, 340, 386, 1058, 1063, 1092, 1112, 1113, 1181, 1182, 1197, 1308, 1401, and 1233	• The control of corruption index measures the "extent to which public power is exercised for private gain, including both petty and grand forms of corruption" [−2.5 to 2.5], higher number represents a lower level of corruption • The Corruption Perceptions Index measures the severity of public sector corruption in a country [0 to 100, lower numbers represent a higher level of corruption] • Survey questions from World Happiness Report: (1) "Is corruption widespread throughout the government or not"; (2) "Is corruption widespread within business or not"	• Subjective survey questions • The Life In Transition Survey (LITS)	103, 110, 113, 148, 211, 166, 361, 1004, 1046, 1067, 1068, 1208, 1219, and 1292	• "To what degree is corruption an obstacle to the current operations of this establishment" [0 = "no obstacle", 4 = very severe obstacle] (WBES) • "Would this establishment's total annual costs increase, remain the same or decrease over the next fiscal year if corruption is no longer an obstacle?" [1 = "increase", 2 = "remain the same", 3 = "decrease"] (WBES) • "It is common for firms in my line of business to have to pay some irregular 'additional payments or gifts' to get things done with regard to customs, taxes, licenses, regulations, services etc." [1 = "never", 6 = "always"] (WBES) • The percentage of firms expected to make informal payments to "get things done" (WBES)
Objective	• Firm-level survey questions aggregated region/ industry/country level • Corruption convictions index • Corruption statistics from The Procuratorial Year Book (China) • Newspaper articles related corruption • Capture index (Russia)	12, 83, 84, 135, 288, 342, 550, 1002, 1005, 1019, 1028, 1051, 1058, 1077, 1083, 1092, 1159, and 1038	• Number of corruption convictions of public officials in each state (per 100,000 capita) (US Department of Justice) • Registered cases of corruption and bribery across provinces in in China (per 100,000 people) (The Procuratorial Year Book) • The number of articles in provincial CPC (China Communist Party) newspapers that promote anti-corruption efforts and denounce corruption. • The Liberal Democracy Index(LDI) (Varieties of Democracy Project)	• Objective survey questions • Survey of Small and Medium Scale Manufacturing Enterprises (SMEs) in Vietnam • Provincial Competitiveness Index (PCI), a nationwide survey of private firms in Vietnam • The entertainment and travel expense disclosed in a firm's annual report. • Survey on China's Private Enterprises	20, 22, 41, 59, 67, 109, 135, 226, 227, 552, 1001, 1007, 1012, 1032, 1046, 1079, 1231, 1038, 1094, and 1154	• "On average what percentage of total annual sales, or estimated total annual value, do establishments like this one pay in informal payments or gifts to public officials for this purpose?" [0-100] (WBES) • Entertainment and travel expenses disclosed in the annual reports of Chinese firms. Excessive expenditures are computed by regressing entertainment and travel expenses against standard operational costs and taking the residual • 4-item measure that gauges a firm's commitment to engaging in anti-corruption practices • 3-item measure based on frequency that firms make informal payments for (1) getting things done, (2) dealing with courts, and (3) dealing with taxes and tax collection

Appendix F. Innovation measurements

	Aggregate-level (country, region, or industry)		Firm-level	
	Sources	Description/examples	Sources	Description/examples
Input	• Global Innovation Index (GII) • World Economic Forum's executive opinion survey • World Development Indicators and CANA panel dataset (Mthanti and Ojah, 2018)	• GII is a composite of measures of innovation inputs and outputs collected from external sources such as the WB and WEF • WEF's executive opinion survey is a composite of items that measure capacity for innovation, quality of scientific institutions, company R&D expenditures, university and industry collaboration, government procurement of technology, and availability of scientists and engineers	• EU Industrial R&D Investment Scoreboard database • DataStream • Amadeus database • Worldwide Patent Statistical database • WBES • China Stock Market and Accounting Research Database	• Elasticity of R&D investments in sales per workers (firms) • R&D over sales • Log of R&D expenditure over fixed assets • Patents granted • patent citations • R&D investment • Binary indicator of R&D • log of R&D expenditure

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	Aggregate-level (country, region, or industry)		Firm-level	
	Sources	Description/examples	Sources	Description/examples
		• Mthanti and Ojah, 2018 construct a composite measure of innovativeness based on investments in R&D, the number of scientific and technical journal articles published per million people and US patents granted per country of origin	• Survey on China's Private Enterprises • India Public Enterprise Survey	• Binary indicator of acquisition of external knowledge (patents, non-patented inventions, and other types of knowledge) • Binary indicator of intangible assets acquired for the purpose of innovation (e.g. trademarks, copyrights, patents, licenses)
Output	• Global Innovation Index (GII) • World Intellectual Property Organization (WIPO) patent data • United States Intellectual Property and Trademark organization (USPTO) patent data • World Bank DataBank • World Bank Development Indicators • State Intellectual Property Office of China • Eurostat regional statistics patent data • OECD regional database • Federal State Statistics Bureau of the Russian Federation (282) • SCOPUS (340) • Nanotechnology product database • Global Entrepreneurship Monitor	• Number of patents or patents per capita • Natural log + 1 of number of patents • Number of utility patents or utility patents per capita • Number of design patents or design patents per capita • Number of patent citations • Number of patents from energy industries • Number of patent applications • Number of trademark applications • Number of articles in scientific journals • Number of peer-reviewed articles on Nanoscience/ Nanotechnology • Number of products listed in the nanotechnology product database • High-tech exports as a % of manufactured exports	• China Stock Market & Accounting Research Database (CSMAR) • Kogan et al., (2017) (original source: Google Patents) • European Patent Office World Patent Statistical Database • Survey of Small and Medium Scale Manufacturing Enterprises (SMEs) in Vietnam • WBES	• Patent applications • Patents granted • Patent citations • Environmental certification obtained • Dichotomous answer to whether a firm applied for a patent over a period of time • Dichotomous answer to whether a firm engaged in product innovation • Dichotomous answer to whether a firm engaged in process innovation • Dichotomous answer to whether a firm engaged in marketing innovation • Dichotomous answer to whether a firm engaged in organizational innovation

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